

The Decay Envelope of the Rvachev Sinc Product

A comparative audit of four proposed solutions, with numerical adjudication and a corrected constant in the published literature

Comparative analysis for Vladimir Reshetnikov’s ProveIt project
covering the four drafts in `drafts/rvachev_up_fourier_decay/`

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Abstract

Mathematics Stack Exchange question 2745497 asks for a positive, analytic, monotone decreasing gauge g such that the Cesàro mean of $|f|/g$ tends to one, where $f(x) = \prod_{n \geq 0} \text{sinc}(\pi x/2^n)$ is the Fourier transform of Rvachev’s up-function. Four AI-produced documents in this directory propose solutions. This audit verifies their overlapping claims numerically and against the literature, adjudicates their one genuine mutual contradiction, and states what the answer to the original question actually is.

The four documents agree, correctly, on the universal factor $\exp(-(\log x)^2/(2 \log 2))$ and on a spectrum of power corrections $x^{-\kappa}$ that depends on the meaning of “amplitude”: $\kappa_\infty = 2.3590148791 \dots$ for the extremal envelope, $\kappa_2 = \log_2(2\pi) = 2.6514961295 \dots$ for the quadratic mean, $\kappa_1 = 2.7480700149 \dots$ for the arithmetic (Cesàro) mean, and $\kappa_0 = 3.1514961295 \dots$ for Lebesgue-typical points. Documents 3 and 4 answer the literal question: the gauge $g(x) = A_1 x^{-\kappa_1} \exp(-(\log x)^2/(2 \log 2))$ with $\kappa_1 = \frac{1}{2} + \log_2(\pi/\varrho_1) = 2.748070014871334 \dots$, where $\varrho_1 = 0.661322602060565 \dots$ is the Perron root of an explicit transfer operator, satisfies the two-sided bounded form of the question and attains the Cesàro limit 1 along $t = 2^k$; but we prove that no gauge whose within-shell shape stabilizes—in particular no Hardy-field gauge—can attain the limit along all t : the limit points trace a nonconstant dyadic-mantissa profile (grid-resolved here to $[0.9249, 1.1133]$; finer quadrature at $y = 8/7$ and $10/7$ finds narrow excursions reaching $\approx [0.9163, 1.1181]$ —see the second-wave postscript), and constancy of the profile would force the transfer-operator eigenmeasure to be absolutely continuous, contradicting its singularity, which we prove from the elementary bound $\varrho_1 \geq 3^{-1/2} > \frac{1}{2}$. The second-wave documents later proved that a deliberately *shell-adaptive* analytic decreasing gauge does attain the unrestricted limit, so the impossibility is a class restriction, not an absolute one (section 13). We add one repair, with proof: with the same gauge, the *logarithmic* mean $\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|f(x)|}{g(x)} \frac{dx}{x}$ converges for all $t \rightarrow \infty$, so the multiplicative form of the question has an exact affirmative answer.

The audit’s principal forensic finding concerns the law of the iterated logarithm. Document 1 derives, by a martingale argument, per-step variance $\pi^2/4$ and hence the fluctuation constant $\pi/\sqrt{2}$; Document 3 quotes the published Theorem 1.2 of Aistleitner–Hofer–Larcher (Ann. Inst. Fourier 67 (2017) 637–687), whose constant is π . Both cannot be right. An exact Parseval computation of the paper’s own variance functional (2.9)—confirmed by measurement over exact rational orbits, $\text{Var}(S_{1000}) = 2459.5$ against the predictions 2464.1 for variance $\pi^2/4$ and 4928.2 for variance $\pi^2/2$ —shows that Document 1 is correct and that the published equation (4.5) omits the Parseval factor $\frac{1}{2}$: the constants in the published Theorems 1.1 and 1.2 should each be divided by $\sqrt{2}$.

Two interpretations are developed beyond what any of the four documents attempts. The envelope-of-lobe-maxima reading (section 6): the peak sequence E_n reproduces the decay spectrum in miniature—limsup envelope κ_∞ , zero liminf on that scale, Cesàro exponent κ_1 via a new peak–integral comparison lemma (with the measured rigidity $\sum_{\text{shell}} E_n / \int_{\text{shell}} |f| = 1.9237$, stable to five digits across six octaves), and a density-one κ_0 law with a central limit theorem whose finite-size right-tail truncation is explained quantitatively by the Gelfond cap. The root-mean-square reading (section 7): the $q = 2$ layer is exactly solvable—the transfer operator has the

closed subleading eigenvalue $-\frac{1}{4}$ with eigenfunctions $\sin 2\pi x$ and $1 + 3 \cos 2\pi x$, making the RMS envelope constant $A_2 = 0.103008918$ converge at the alternating geometric rate $(-\frac{1}{2})^k$ —yet the all- t Cesàro limit still fails (singular eigenmeasure, profile $[0.894, 1.192]$), and the total energy check $\int_{\mathbb{R}} f^2 = \frac{1}{2} + \sum_m f(m + \frac{1}{2})^2 = 0.808873535967972$ closes to $2 \cdot 10^{-15}$.

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1 Purpose, inputs, and method

This document audits the four solution drafts

label	file
Document 1	rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
Document 2	Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
Document 3	Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
Document 4	rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex

which respond to the problem posed in the adjacent OCR copy of Mathematics Stack Exchange question 2745497 [6] and, indirectly, to the now-deleted repository draft *Decay rate conjecture* (git object e1cdd6260, quoted in [section 11.1](#)).

The audit method follows the repository’s engineering policy, with one governing principle: **every adjudication in this document rests on a proof; numerical experiments serve only as insight and as validation of conclusions reached rigorously**. Where a verdict is stated ([theorems 5.5, 5.7, 6.2, 7.2, 9.1](#) and [9.2, finding 9.3](#)), a proof or a complete proof architecture is given, with the imported classical inputs named explicitly ([section 5.2](#)). Independently of that, every quantitative sentence was *computed*, not reasoned to: all numerical values were produced by the Python scripts reproduced in [section B](#), run on 26 August 2026, and independent computations (direct quadrature versus spectral prediction, direct maximization versus published tables, Monte Carlo versus closed-form variances) are compared digit by digit in [section 10](#). Literature quotations from Aistleitner–Hofer–Larcher [1] were checked against three independent sources: the publisher’s PDF *as rendered page images* (pages 641, 651, 663), the OCR transcription stored at docs/papers/AIF_2017__67_2_637_0/, and the authors’ own L^AT_EX source of the arXiv version at docs/papers/arXiv-1502.06738v2/Lacuna ry_products.tex. All three agree on every formula quoted here, so no OCR artifact affects the findings.

2 The object and the question

Throughout, $\operatorname{sinc} z = \sin z/z$, $\operatorname{sinc} 0 = 1$, $a = \log 2$, and

$$f(x) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^n}\right), \quad x \in \mathbb{R}. \quad (1)$$

Under the convention $\widehat{h}(\xi) = \int h(t)e^{-2\pi i \xi t} dt$ one has $f = \widehat{\operatorname{up}}$, the Fourier transform of Rvachev’s up-function; in the Lean development this object is `rvachevFourier_eq_product` with refinement law `rvachevFourier_scaling`. The function is entire, even, vanishes at each nonzero integer m with multiplicity $1 + v_2(|m|)$, has exactly one critical point between consecutive integer zeros, and its sign on $(m, m + 1)$ is the signed Thue–Morse value $(-1)^{s_2(m)}$ (all four documents prove these; the sign statement is also in the Thue–Morse Formula Atlas, §“The sign of the Fabius sinc product”). All three background facts are formalized in this same normalization, and are cited here rather than reproved. *Multiplicity*: the dyadic zero lattice meets the integer $m \geq 1$ exactly $1 + v_2(m)$ times (`LatticeFiber`) and $\|\Phi(x)\|/|x - m|^{1+v_2(m)}$ tends to an explicit positive constant along the punctured neighbourhood of m (`PhiZeroOrder`, `tendsto_norm_div_pow_zero_order`), with the odd and 2^j extremes as corollaries (`ValleyZeroOrders`); the Lean statements are for $m \geq 1$, the negative side following by evenness. *One peak*: $\log \|\Phi\|$ is strictly concave on every lobe (`OnePeakPerLobe`), which with continuity of Φ on the real axis gives *exactly one* maximizer of $|\Phi|$ per side lobe (`RvachevProductContinuity`, `existsUnique_isMaxOn_lobe`) and the central peak exactly at the origin (`CentralLobePeakAtZero`) — the equivalent maximizer form of “one critical point per lobe”, which is what the library states. *Sign*: `rvachevFourierProduct_eq_thu`

`eMorse_sign_mul_norm` (`ThueMorseLobeSign`), built on `PhiRealSign`, `LobeSignCount` and `LobeSignSelfSimilarity`.

The question [6] asks for a real g , “elementary, if possible ... analytic, positive and monotone decreasing (and having monotone derivatives of any order, if possible)”, such that

$$\lim_{t \rightarrow \infty} \frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx = 1 \quad (\text{Q2})$$

for large enough t_0 , or at least

$$\exists 0 < A < B \forall t > t_0 : \quad A < \frac{1}{t - t_0} \int_{t_0}^t \frac{|f(x)|}{g(x)} dx < B. \quad (\text{Q3})$$

The asker’s empirical guess was “something close to $\exp(-\log^2 x)$, but perhaps not exactly that”.

Two remarks on reading the question. First, the Cesàro mean in (Q2)–(Q3) makes this an L^1 (arithmetic-mean) question: g is asked to be the *mean local amplitude* of $|f|$, not a pointwise envelope, not an envelope of the lobe maxima, and not a root-mean-square amplitude. Much of what the four documents prove concerns those other readings; [section 3](#) maps out how the answer changes with the reading. Second, since $\int_{t/2}^t$ carries half the weight of $\int_{t_0}^t$, condition (Q2) is equivalent to the statement that the *partial-shell* averages $\frac{1}{2^k} \int_{2^k}^{2^{k+1}} |f|/g dx \rightarrow y - 1$ for every $y \in [1, 2]$; this equivalence is what ultimately decides (Q2) negatively for natural gauges ([section 5](#)).

3 What “asymptotic decay rate” can mean

All four documents, correctly, refuse to assign the product a single decay rate. After the universal factor

$$\mathcal{E}(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2}\right),$$

whose exponent’s coefficient $-1/(2 \log 2) = -0.7213475 \dots$ comes from the deterministic denominator $\prod_{j \leq \log_2 x} 2^j$, the surviving power of x depends on the interpretation. The verified dictionary:

Interpretation	power κ after $\mathcal{E}(x)$	status
pointwise equivalent $ f \sim g$	none	impossible: integer zeros; and on a.e. dyadic ray the centered logarithm has unbounded iterated-logarithm excursions
envelope of <i>all</i> lobe maxima E_n	none	impossible even after passing to local maxima: the peaks straight after 2^k decay with the doubled quadratic coefficient $-1/\log 2$ (Docs. 2, 4; verified section 10.5); full four-layer analysis in section 6
largest lobe in $[2^k, 2^{k+1}]$	$\kappa_\infty =$ 2.3590148791	sharp as a limsup; the true annular maximum carries a second log-normal factor $\exp(-(\log \log x)^2/(2 \log 2) + \dots)$ (Docs. 1, 2; verified section 10.4)
Lebesgue-typical dyadic ray	$\kappa_0 =$ 3.1514961295	holds a.e. with $o(\log x)$ error; the error is genuinely random, of iterated-logarithm size (Docs. 1–4; constant of section 9)
arithmetic mean = the literal question	$\kappa_1 =$ 2.7480700149	(Q3) solvable; (Q2) solvable along $t = 2^k$ and for the logarithmic mean, not for all t (Docs. 3, 4; section 5)
root mean square	$\kappa_2 = \log_2(2\pi) =$ 2.6514961295	exact identity $\int_0^1 \prod_{j < n} \sin^2(\pi 2^j t) dt = 2^{-n}$; this is precisely the exponent of the deleted draft (Docs. 3, 4); exactly solvable layer, developed in section 7

Document 4 embeds the middle rows in a full L^q spectrum $\kappa_q = (\log \pi + \frac{a}{2} - \log \Lambda_q)/a$ with $\Lambda_q = \varrho_q^{1/q}$ the q -th root of the Perron root of the weighted doubling transfer operator; κ_q decreases strictly from κ_0 to κ_∞ . The verified values are collected in [section A](#).

The dictionary map itself is formalized. `KappaDictionary` defines $\kappa(\Lambda) = \frac{1}{2} + \log_2(\pi/\Lambda)$ (`kappaOf`, `kappaOf_eq`), proves it strictly decreasing in the per-shell mean Λ (`kappaOf_strictAntion`), and evaluates it at the three closed-form means $\Lambda = \frac{1}{2}, \sqrt{2}/2, \sqrt{3}/2$ — that is, at $\kappa_0, \kappa_2, \kappa_\infty$ (`kappaOf_half`, `kappaOf_sqrt_two`, `kappaOf_sqrt_three`), together with the exact gaps. Since `Fabius.log_pi_bracket` became available, all three also carry numerical enclosures, in `KappaConstantEnclosures`:

$$\begin{aligned} 3.1514961287 &\leq \kappa_0 \leq 3.1514961300, \\ 2.6514961287 &\leq \kappa_2 \leq 2.6514961300, \\ 2.3590148784 &\leq \kappa_\infty \leq 2.3590148796, \end{aligned}$$

(`Fabius.kappa_zero_bracket`, `Fabius.kappa_two_bracket`, `Fabius.kappa_inf_bracket`), of widths 1.3, 1.3 and $1.2 \cdot 10^{-9}$, so seven decimals of κ_0 and κ_2 and eight of κ_∞ are kernel-checked — the extra decimal being where the values fall against a decimal boundary, not extra precision. Because κ_0 and κ_2 differ from $\log_2 \pi$ by an additive rational, that one quantity is bracketed once (`Fabius.logTwo_pi_bracket`) and both are read off with no further loss. These use `Mathlib`'s `log 2` and `log 3` in their tightest form, `Real.log_two_near_10` and `Real.log_three_near_10`, rather than the `_d9` decimal corollaries; what dominates the residual width is $\log \pi$, whose own $5 \cdot 10^{-10}$ width is inherited from the `_d9` form of `log 3` used to prove it. The κ_1 row is deliberately absent from the dictionary: the Perron root exists in the library only existentially, so every Lean statement about κ_1 carries it as a hypothesis (see `KappaOneEnclosure`, below). What the dictionary does not supply is the *attribution* of each κ to its interpretation; that is formal so far only for κ_∞ , pinned from both sides by `GlobalDecayEnvelope` and `PeakRayEnvelope`, and, in measure rather than almost everywhere, for κ_0 (`KappaZeroInMeasure`, `FluctuationTightness`).

Two further interpretation issues are settled by the problem statement itself. The convention $\text{sinc } z = \sin z/z$ (argument already containing π) is forced by the stated zero set $\mathbb{Z}_{>0}$ and by the plot in the question. And the normalization matters only through trivial rescalings: $f(x) = \widehat{\text{up}}(x)$ exactly in the $e^{-2\pi i x t}$ convention, so every statement transfers verbatim to the up-function literature.

Finally, the two forms (Q2) and (Q3) are genuinely different problems: (Q3) is solved by an explicit elementary gauge, while (Q2) fails for every gauge in the natural class ([theorem 5.1](#)). The choice of *averaging operator* is thus itself an interpretation with mathematical content: arithmetic averaging does not respect the multiplicative self-similarity $f(2x) = \text{sinc}(2\pi x)f(x)$, and the obstruction to (Q2) disappears the moment the average is taken multiplicatively.

4 The common core, verified

All four documents rest on the same exact identity, stated here in the normalization of Document 3. Write $x = 2^k y$, $1 \leq y < 2$, $t = y - 1$, $n = k + 1$,

$$P_n(t) = \prod_{j=0}^{n-1} |\sin(\pi 2^j t)|, \quad H(y) = \prod_{m=1}^{\infty} \left| \text{sinc} \frac{\pi y}{2^m} \right|.$$

Then

$$|f(2^k y)| = 2^{-k(k+1)/2} (\pi y)^{-(k+1)} H(y) P_{k+1}(y-1). \quad (2)$$

Numerical check: for $k \in \{3, 7, 12\}$ and random y , the two sides of (2) agree to $1.6 \cdot 10^{-12}$ in the logarithm (dominated by quadrature-free floating-point accumulation). The identity reduces every

averaged decay question to the lacunary product P_n under the doubling map $Tt = 2t \bmod 1$, and the elementary algebra

$$-\frac{a}{2}k(k+1) - (k+1)(c + \log y) = -\frac{L^2}{2a} - \left(\frac{1}{2} + \frac{c}{a}\right)L + \Phi_c(\log y), \quad L = \log x,$$

(Document 3's identity, reproved symbolically) converts each per-level numerator scale e^{-c} into the power $\kappa = \frac{1}{2} + c/a$. The problem statement's own finite expansion

$$\prod_{n=0}^m \operatorname{sinc} \frac{\pi x}{2^n} = \frac{2^{\binom{m}{2}}}{(\pi x)^{m+1}} \sum_{n=0}^{2^m-1} t_n \sin\left(\frac{\pi m}{2} + \frac{2n+1}{2^m} \pi x\right)$$

was verified symbolically for $m = 0, 1, 2$; it is the Thue–Morse sine-product identity of the Formula Atlas (§“Sine and cosine products as Thue–Morse sums”) and is the bridge to Gelfond-type estimates: $2^n P_n(t)$ is the modulus of the Thue–Morse trigonometric polynomial $\sum_{r < 2^n} (-1)^{s_2(r)} e^{2\pi i r t}$. The identity is not merely spot-checked but proved: it is `LacunaryProductToSum's` `prod_sin_two_pow_eq_thueMorse_sum`, $2^m \prod_{j \leq m} \sin(2^j w) = \sum_{n < 2^m} \varepsilon_n \sin(m\pi/2 + (2n+1)w)$, for every m and every real w .

5 The answer to the original question

This section states the audited answer, synthesizing Documents 3 and 4 (who give it) with the numerical realization computed for this audit and one new repair.

Let $\mathcal{L}_1$ be the weighted doubling transfer operator

$$(\mathcal{L}_1 h)(x) = \frac{1}{2} \left[\sin\left(\frac{\pi x}{2}\right) h\left(\frac{x}{2}\right) + \cos\left(\frac{\pi x}{2}\right) h\left(\frac{x+1}{2}\right) \right], \quad 0 \leq x \leq 1, \quad (3)$$

so that $\int_0^1 P_n(t) q(t) dt = \int_0^1 (\mathcal{L}_1^n q)(x) dx$, and let ϱ_1 be its Perron root. Then:

$$\begin{aligned} \varrho_1 &= 0.661322602060565 \dots && (\text{collocation, } N = 16\text{--}40, \text{ stable to } 2 \cdot 10^{-15}) \\ \kappa_1 &= \frac{\log \pi + \frac{a}{2} - \log \varrho_1}{a} = \frac{1}{2} + \log_2 \frac{\pi}{\varrho_1} = 2.748070014871334 \dots \\ g_1(x) &= x^{-\kappa_1} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad A_1 = 0.09126612 \dots \end{aligned} \quad (4)$$

where A_1 is the limit of the dyadic-shell means of $|f|/g_1$ (section 10.1). The asymptotic $I_1(n) := \int_0^1 P_n \rightarrow \kappa \lambda^n$ with $\lambda = \varrho_1$ is due to Fouvry–Mauduit [3]; the enclosure $0.66130 < \lambda < 0.66135$ is Lemma 5.1 of [1]; Document 4's exact-arithmetic Collatz–Wielandt certificate, which this audit executed verbatim (section 10.7), proves $0.66126807 < \varrho_1 < 0.66134891$ by Sturm sequences alone; and the kernel-checked Lean enclosure $0.66126798 \leq \varrho_1 \leq 0.66134921$ certifies the same test function through subdivided Bernstein certificates (`PerronRootEnclosure`, section 10.7).

Theorem 5.1 (audited answer to [6]). *With $g = A_1 g_1$ as in (4):*

1. **(Q3) holds.** *The Cesàro ratio is eventually confined to a compact subinterval of $(0, \infty)$; its set of limit points is exactly the range of the continuous dyadic-mantissa profile $\mathcal{P}$ of item (iii), numerically containing $[0.91627, 1.11814]$ (the step-0.025 grid of figure 1 resolves only $[0.9249, 1.1133]$; see section 13). The gauge is real-analytic and positive on $(0, \infty)$, eventually strictly decreasing, and each fixed derivative order is eventually monotone with alternating sign, meeting every regularity requested in the question.*
2. **(Q2) holds along dyadic times.** *$\frac{1}{t-t_0} \int_{t_0}^t |f|/g \rightarrow 1$ as $t \rightarrow \infty$ through $t = 2^k$ (measured: 0.999992 at $k = 5$, 1.000000 for $k \geq 9$).*

3. (Q2) fails for all t , for every natural gauge (theorem 5.5). For $t = 2^k y$, $y \in [1, 2]$ fixed, the Cesàro ratio converges to

$$\mathcal{P}(y) = \frac{1 + \mathcal{F}(y - 1)}{y}, \quad \mathcal{F}(z) = \lim_{n \rightarrow \infty} \frac{\int_0^z W(t) P_n(t) dt}{\int_0^1 W(t) P_n(t) dt},$$

with W the bounded mantissa weight of the exact factorization. The limit distribution $\mathcal{F}$ is the (normalized, W -weighted) eigenmeasure of $\mathcal{L}_1$; it is continuous and singular, so $\mathcal{F}(z) \neq z$ and $\mathcal{P} \neq 1$. Consequently no gauge whose within-shell shape stabilizes—in particular no Hardy-field gauge, with or without a continuous log-periodic modulation—can satisfy (Q2); the second wave strengthened this to all bounded-distortion gauges, and conversely constructed a shell-adaptive analytic gauge that does attain (Q2) (section 13). Measured profile: grid minimum 0.9249 near $y = 1.15$, maximum 1.1132 near $y = 1.43$ (figure 1); narrow excursions near $y = 8/7$ and $10/7$ reach 0.91627 and 1.11814.

4. **Repair (new here): the multiplicative form of (Q2) is solvable exactly** (theorem 5.7). With the same g_1 and the constant $A_1^{\log} = 0.0938606357568 \dots$ (corrected here; the value 0.0938610 printed in the first edition of this audit was a finite-shell running mean, wrong in its seventh figure — section 13),

$$\lim_{t \rightarrow \infty} \frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|f(x)|}{A_1^{\log} g_1(x)} \frac{dx}{x} = 1 \quad \text{for all } t \rightarrow \infty.$$

Measured: all four tracked mantissas converge to the same constant, with the expected $\mathcal{O}(1/\log t)$ speed (section 10.3).

5.1 Rigorous basis

The positive parts (i)–(ii) are Theorem 6.4 of Document 3 and Theorem 5.3/Corollary 6.2 of Document 4; their proofs were audited line by line and rest on the exact factorization plus the Fouvry–Mauduit asymptotic and the Ruelle–Perron–Frobenius (RPF) convergence for $\mathcal{L}_1$ (see section 5.2 for the exact division between proved and imported). The negative part (iii) and the repair (iv) are proved here in full, since they are the audit’s verdicts on the question itself.

Formal status. The purely limit-theoretic cores of theorem 5.1 are kernel-checked in the Lean development, each stated for an arbitrary cumulative mass function with the RPF input left as a hypothesis; they therefore make the theorem *conditionally* formal, and are not the same thing as the statements above being formalized. Item (i) is BoundednessSkeleton’s eventually_bounded_of_monotone_of_dyadic: a monotone mass G with the dyadic normalization $G(2^K)/2^K \rightarrow A > 0$ satisfies $A/4 \leq G(t)/t \leq 3A$ for all large t — a two-sided bracket, weaker than the sharp compact limit set asserted in (i). The shell arithmetic of (iii) is CesaroProfileSkeleton’s tendsto_cumulative_profile: given the completed-shell limit A and the terminal partial-shell limit P at mantissa y , the Cesàro ratio along $t = 2^k y$ converges to $(A + P)/y$ — exactly the profile $\mathcal{P}$ whose nonconstancy is the obstruction, with the aggregation step supplied by GeometricCesaro (tendsto_geom_shell_average, itself a corollary of the general weighted Toeplitz theorem tendsto_weighted_average). Item (iv) is LogMeanSkeleton’s tendsto_sum_add_bdd_div. What none of them contains is precisely the imported block of section 5.2: the RPF convergence that supplies the shell limits, and with it the singularity of ν that makes the profile nonconstant. The library’s Perron layer stops short of that — existence of the root $\varrho_1 \in [\frac{1}{2}, \sqrt{2}/2]$ by Fekete (PerronRootExistence) and the Lasota–Yorke pair for $\mathcal{L}_1$ (TransferOperatorStep, LasotaYorkeIteration) are formal, the spectral gap and the eigenfunction are not.

Lemma 5.2 (reduction of (Q2) to partial-shell linearity). *Let $g > 0$ be measurable and $F(t) = \int_{t_0}^t |f|/g$. If (Q2) holds, then for every $y \in [1, 2]$,*

$$\frac{F(2^k y) - F(2^k)}{2^k} \rightarrow y - 1 \quad (k \rightarrow \infty). \quad (5)$$

Proof. (Q2) says $F(t) = (t - t_0)(1 + o(1))$. Insert $t = 2^k y$ and $t = 2^k$ and subtract: $F(2^k y) - F(2^k) = 2^k(y - 1 + o(1))$, the error uniform for $y \in [1, 2]$ since y is bounded. $\square$

Lemma 5.2 is formalized, and in greater generality: `ShellLinearity's tendsto_shell_increment` proves $(F(2^k(1+z)) - F(2^k))/2^k \rightarrow z$ for an arbitrary $F : \mathbb{R} \rightarrow \mathbb{R}$ and every $z \geq 0$, not only for $y = 1 + z \in [1, 2]$, with the ray normalization $F(2^k c)/2^k \rightarrow c$ as `tendsto_ray_div_two_pow`. The uniformity in y noted in the proof above is not part of the Lean statement, which fixes z and takes the limit in k .

Proposition 5.3 (elementary two-sided bracket for ϱ_1). *Let $I_1(n) = \int_0^1 P_n$. Then for all $n \geq 1$,*

$$\frac{3^{-n/2}}{\sqrt{5/3}} \leq I_1(n) \leq \sqrt{\frac{5}{3}} \left(\frac{\sqrt{3}}{2}\right)^n,$$

so any limit value $\varrho_1 = \lim I_1(n)^{1/n}$ satisfies $3^{-1/2} \leq \varrho_1 \leq \sqrt{3}/2$; in particular $\varrho_1 > \frac{1}{2}$, hence $\log(2\varrho_1) > 0$.

Proof. Upper bound: Document 2's subaction telescope (audited, purely algebraic; Document 4's two-step inequality gives the slightly sharper constant $2/\sqrt{3}$) yields the pointwise Gelfond bound $P_n \leq \sqrt{5/3} (\sqrt{3}/2)^n$; integrate. Lower bound: by Cauchy-Schwarz and the exact identity $\int_0^1 P_n^2 = 2^{-n}$ (proposition 5.4),

$$2^{-n} = \int_0^1 P_n \cdot P_n \leq \|P_n\|_\infty \int_0^1 P_n \leq \sqrt{\frac{5}{3}} \left(\frac{\sqrt{3}}{2}\right)^n I_1(n),$$

whence $I_1(n) \geq (5/3)^{-1/2} (2 \cdot \sqrt{3}/2)^{-n} = (5/3)^{-1/2} 3^{-n/2}$. (The certificate of section 10.7 pins $\varrho_1 \in (0.66126807, 0.66134891)$, far inside this bracket, but the soft bound $\varrho_1 \geq 3^{-1/2} = 0.577\dots$ is all that theorem 5.5 needs, and it is fully elementary.) $\square$

Both halves of proposition 5.3 are formalized. The lower half is `le_mul_integral_prod_abs_sin_two_pow` (`GelfondLogisticBound`), which is the statement above; the upper half is `SincMeanBracket's integral_prod_abs_sin_two_pow_le`, which proves the sharper $I_1(n+1) \leq (\sqrt{3}/2)^n$ — its pointwise input being the sharp uniform bound `abs_prod_sin_two_pow_le_sharp` (`SharpGelfondBound`) rather than the $\sqrt{5/3}$ -constant bound used here. So the Lean bracket $3^{-1/2} \leq \liminf I_1(n)^{1/n} \leq \limsup I_1(n)^{1/n} \leq \sqrt{3}/2$ is this proposition's conclusion, reached by a slightly better route.

Proposition 5.4 (exact L^2 identity). $\int_0^1 P_n(t)^2 dt = 2^{-n}$ for every $n \geq 1$.

Proof. $P_n^2 = \prod_{j < n} \sin^2(\pi 2^j t) = 2^{-n} \prod_{j < n} (1 - \cos(2\pi 2^j t))$. Expanding the product gives $\pm$ products of cosines with frequencies $\sum_{j \in S} 2^j$ over nonempty $S \subseteq \{0, \dots, n-1\}$ after linearization; by uniqueness of binary representations no cancellation to frequency zero occurs, so only the constant term 1 survives integration. $\square$

Formalized, and more generally than stated: `LacunaryRieszIntegral` proves the master orthogonality theorem `integral_cos_mul_rieszProduct` for arbitrary Hadamard-gap integer frequencies, amplitudes and phases, with proposition 5.4 the corollary `integral_prod_sin_sq_two_pow` — valid there for every $n \geq 0$ — and `integral_prod_sin_sq_pow` the same identity for an arbitrary integer base $b \geq 2$.

Theorem 5.5 (impossibility of (Q2) in the stabilizing class). *Let $g > 0$ be a gauge whose within-shell shape stabilizes: there are constants $c_k > 0$ and a continuous $u : [0, 1] \rightarrow (0, \infty)$ with*

$$\frac{g(2^k(1+z))}{c_k g_1(2^k(1+z))} \longrightarrow u(z) \quad \text{uniformly in } z \in [0, 1].$$

(The class contains every Hardy-field gauge of the correct rough size, and every such gauge times a continuous positive function of $\{\log_2 x\}$.) Then (Q2) fails for g .

Proof. Suppose (Q2) holds. By the exact factorization (2) and the κ -algebra, on the shell $[2^k, 2^{k+1}]$,

$$\frac{|f(2^k(1+z))|}{g_1(2^k(1+z))} = \varrho_1^{-(k+1)} W(z) P_{k+1}(z),$$

where $W(z) = H(1+z)(1+z)^{\kappa_1-1} \exp(\log^2(1+z)/(2a)) \cdot \pi^{-1} \varrho_1$ collects the bounded mantissa factors; W is continuous on $[0, 1]$ and strictly positive on $[0, 1)$. Hence, with $q = u^{-1}W$ (continuous, positive on $[0, 1)$),

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} = \frac{1+o(1)}{c_k \varrho_1^{k+1}} \int_{z_1}^{z_2} q(z) P_{k+1}(z) dz.$$

By the RPF convergence for $\mathcal{L}_1$ (section 5.2), for every continuous ϕ , $\varrho_1^{-n} \int_0^1 \phi P_n \rightarrow \nu(\phi) \int_0^1 h$, where $h > 0$ is the eigenfunction and ν the eigenmeasure, normalized by $\nu(h) = 1$; and ν has no atoms (Document 4's argument, audited: $\varrho_1 \nu(\{z\}) = \frac{1}{2} s(z) \nu(\{Tz\})$ forces any atom's forward masses to grow by factors $\geq 2\varrho_1 > 1$ by proposition 5.3, contradicting finiteness). Atomlessness upgrades the convergence from continuous ϕ to indicators of intervals, so

$$\frac{F(2^k(1+z_2)) - F(2^k(1+z_1))}{2^k} \rightarrow C \nu(q \mathbf{1}_{[z_1, z_2]}), \quad C = \lim \frac{\int_0^1 h}{c_k \varrho_1^{k+1}} \in (0, \infty), \quad (6)$$

the existence and positivity of C being forced by lemma 5.2 with $(z_1, z_2) = (0, 1)$. Comparing (6) with (5) gives $C \nu(q \mathbf{1}_{[0, z]}) = z$ for all $z \in [0, 1]$: the finite Borel measure $Cq d\nu$ is Lebesgue measure on $[0, 1]$. Since q is continuous and positive on $[0, 1)$, this forces $\nu \upharpoonright_{[0, 1)} \ll \text{Leb}$, and $\nu(\{1\}) = 0$ by atomlessness, so $\nu \ll \text{Leb}$.

But ν is singular. Indeed, if $d\nu = r d\text{Leb}$ with $0 \leq r \in L^1$ not a.e. zero, the eigenmeasure equation $\mathcal{L}_1^* \nu = \varrho_1 \nu$ unwinds, by the change of variables on the two branches, to the a.e. conformal relation $s(z) r(Tz) = \varrho_1 r(z)$, $s(z) = |\sin \pi z|$. Iterating,

$$r(T^n z) = \frac{\varrho_1^n}{P_n(z)} r(z),$$

and by the Birkhoff ergodic theorem $P_n(z) = \exp(-n(\log 2 + o(1)))$ for Lebesgue-a.e. z (the potential $\log s$ is integrable with mean $-\log 2$), so on $\{r > 0\}$, a set of positive measure, $r(T^n z) = r(z) \exp(n(\log(2\varrho_1) + o(1))) \rightarrow \infty$ geometrically, using $\log(2\varrho_1) > 0$ from proposition 5.3. On the other hand T preserves Lebesgue measure, so for any $\varepsilon > 0$, $\text{Leb}(r \circ T^n > e^{\varepsilon n}) = \text{Leb}(r > e^{\varepsilon n}) \leq \|r\|_1 e^{-\varepsilon n}$, which is summable; by Borel–Cantelli $r(T^n z) \leq e^{\varepsilon n}$ eventually for a.e. z . Choosing $\varepsilon < \log(2\varrho_1)$ yields a contradiction: no such density exists, so ν is *not* absolutely continuous—contradicting $\nu \ll \text{Leb}$ derived above. This refutes (Q2). (The same computation upgrades to full singularity $\nu \perp \text{Leb}$: the dual action $\mathcal{L}_1^* \nu = \sum_b (\tau_b)_* (\frac{1}{2}(s \circ \tau_b) \nu)$ over the two inverse branches τ_b maps absolutely continuous parts to absolutely continuous parts and singular parts to singular parts, so by uniqueness of the Lebesgue decomposition the absolutely continuous part of ν is itself an eigenmeasure, hence zero by the argument just given. This is the precise sense of the singularity statements in Documents 3 and 4.) $\square$

Remark 5.6 (Hardy-field gauges are covered exhaustively). For a Hardy-field gauge g , write $h = \log(g/g_1)$. Membership in a Hardy field makes $xh'(x)$ eventually monotone, so $\gamma = \lim_{x \rightarrow \infty} xh'(x)$ exists in $[-\infty, +\infty]$. If γ is finite, then $h(2^k(1+z)) - h(2^k) \rightarrow \gamma \log(1+z)$ uniformly, the shell shape stabilizes with $u(z) = (1+z)^\gamma$, and theorem 5.5 applies. If $\gamma = +\infty$ (resp. $-\infty$), the factor $e^{-(h(2^k(1+z)) - h(2^k))}$ degenerates monotonically, and—using that the eigenmeasure charges every subinterval (full support, part of the RPF package)—the shell mass of $|f|/g$ concentrates at the left (resp. right) shell edge, so the partial-shell ratio tends to 1 (resp. 0) for every fixed $y \in (1, 2)$, violating the linearity (5) directly. In every case (Q2) fails for every Hardy-field gauge, hence for every gauge given by an elementary closed-form expression.

Theorem 5.7 (the logarithmic mean converges for all t). *There is $A_1^{\log} \in (0, \infty)$ such that, for every fixed $t_0 > 1$,*

$$\frac{1}{\log(t/t_0)} \int_{t_0}^t \frac{|f(x)|}{g_1(x)} \frac{dx}{x} \longrightarrow A_1^{\log} \quad (t \rightarrow \infty, \text{ unrestricted}).$$

Proof. Let $\beta_k = \int_{2^k}^{2^{k+1}} \frac{|f|}{g_1} \frac{dx}{x}$. By the exact factorization, $\beta_k = \varrho_1^{-(k+1)} \int_0^1 \frac{W(z)}{1+z} P_{k+1}(z) dz$, and the RPF convergence applied to the continuous weight $W(z)/(1+z)$ gives

$$\beta_k \longrightarrow \beta_\infty := \nu\left(\frac{W}{1+\cdot}\right) \int_0^1 h \, d\text{Leb} \in (0, \infty);$$

in particular (β_k) is bounded. For $2^K \leq t < 2^{K+1}$,

$$\int_{t_0}^t \frac{|f|}{g_1} \frac{dx}{x} = \sum_{k=k_0}^{K-1} \beta_k + \mathcal{O}(1) + \Theta_K, \quad 0 \leq \Theta_K \leq \beta_K,$$

and $\log(t/t_0) = K \log 2 + \mathcal{O}(1)$. Since $\beta_k \rightarrow \beta_\infty$, Cesàro convergence gives $\frac{1}{K} \sum_{k < K} \beta_k \rightarrow \beta_\infty$, while $\Theta_K/(K \log 2) \rightarrow 0$ because (β_K) is bounded. Hence the logarithmic mean converges to $A_1^{\log} = \beta_\infty / \log 2$, with no constraint on how $t \rightarrow \infty$. The point of the proof is structural: in the logarithmic average the terminal partial shell has weight $\mathcal{O}(1/\log t)$, so the singular within-shell distribution that defeats (Q2) is annihilated rather than merely averaged. $\square$

The bookkeeping of this proof — everything after “ $\beta_k \rightarrow \beta_\infty$ ” — is formalized as `LogMeanSkeleton’s tendsto_sum_add_bdd_div`: if $\beta_k \rightarrow L$ and the terminal partial shell γ and endpoint correction δ are bounded, then $(\sum_{k < K} \beta_k + \gamma_K)/(K + \delta_K) \rightarrow L$. The convergence $\beta_k \rightarrow \beta_\infty$ itself is the RPF input and is not formalized, so [theorem 5.7](#) is formally backed only conditionally.

5.2 Imported inputs, stated as obligations

Per repository policy, the division between what is proved and what is imported:

- *Elementary and fully audited (no external input)*: the factorization (2); the κ -algebra; [propositions 5.3](#) and [5.4](#); the Gelfond bound with constant $\sqrt{5/3}$ (Documents 2 and 4, algebraic identities checked symbolically); the Sturm certificate for ϱ_1 ([section 10.7](#)); atomlessness and singularity of ν ; [lemma 5.2](#); the Cesàro bookkeeping in [theorem 5.7](#); the variance computation of [theorem 9.1](#).
- *Classical theorems used by name*: Birkhoff’s ergodic theorem (typical law; singularity proof); the stationary-ergodic martingale LIL of Stout/Heyde–Scott ([theorem 9.2](#)); the Clausen expansion $\log |2 \sin \pi x| = -\sum_{j \geq 1} \cos(2\pi j x)/j$ in L^2 .
- *Published research input*: the Fouvry–Mauduit asymptotic $I_1(n) = \kappa \varrho_1^n (1 + o(1))$ [3]. It is needed even for the bounded form (Q3) with the exact gauge exponent: the elementary bracket of AHL Lemma 5.1 (whose monotone $m_j \uparrow$, $M_j \downarrow$ argument was audited and is elementary) pins ϱ_1 to five digits but still allows a subexponential drift $I_1(n)/\varrho_1^n = e^{o(n)}$, which a fixed power $x^{-\kappa_1}$ cannot absorb.
- *The one genuinely imported analytic block*: the RPF spectral picture for $\mathcal{L}_1$ (existence of $h > 0$, ν , and $\varrho_1^{-n} \mathcal{L}_1^n \phi \rightarrow h \nu(\phi)$ for continuous ϕ) on a Hölder or bounded-variation space, as cited by Documents 3 and 4 from the standard transfer-operator literature. None of the four documents reproves it, and neither does this audit; it is the standing analytic obligation attached to [theorem 5.1\(ii\),\(iii\)](#) and [theorem 5.7](#). Everything else in this section survives with elementary means alone.

In one sentence. The mean amplitude of $|f|$ is $A_1 x^{-2.7480700149} \exp(-(\log x)^2/(2 \log 2))$ with $A_1 = 0.0912661$; taking g equal to it answers the bounded question (Q3) and the dyadic and logarithmic forms of (Q2), while the unrestricted (Q2) is unattainable for every gauge whose within-shell shape stabilizes or stays within bounded distortion of g : the last dyadic block never forgets the singular way in which the mass of $|f|$ is distributed inside a block. (A deliberately shell-adaptive analytic gauge escapes this, as the second wave later proved: section 13.) The asker’s guess $\exp(-\log^2 x)$ had the right shape; the true quadratic coefficient is $1/(2 \log 2) = 0.7213$, and the missing structure is the power $x^{-\kappa_1}$ whose exponent is fixed by a transfer-operator eigenvalue, not by any classical constant.

Two consistency remarks. First, κ_1 is *provably* strictly between κ_2 and κ_0 : by Cauchy-Schwarz $I_1(n)^2 \leq I_2(n) = 2^{-n}$, so $\varrho_1 \leq 2^{-1/2}$ would be forced the other way—in fact $I_1(n) \geq I_2(n)/\max P_n \geq c(2/\sqrt{3})^{-n}2^{-n}$ gives the compatible ordering $\frac{1}{2} < \varrho_1 < \frac{1}{\sqrt{2}}$, consistent with $0.6613 \in (0.5, 0.7071)$. (The Cauchy-Schwarz half is formalized as `LacunaryMeanSharp’s integral_prod_abs_sin_two_pow_le_sqrt_two`, $I_1(n) \leq (\sqrt{2}/2)^n$, proved there for arbitrary Hadamard-gap integer frequencies and every integer base $b \geq 2$; it is strictly better than the $(\sqrt{3}/2)$ -route upper bound of proposition 5.3.) Second, the deleted draft’s exponent $\log_2(2\pi)$ is exactly κ_2 ; the draft was thus computing the root-mean-square envelope while claiming a pointwise and Cesàro-mean meaning for it. Documents 3 and 4 were the first to make this diagnosis precise, and it is confirmed by the exact identity $I_2(n) = 2^{-n}$ (verified to 10^{-12} for $n \leq 11$; it is a two-line Parseval argument on the Thue-Morse polynomial).

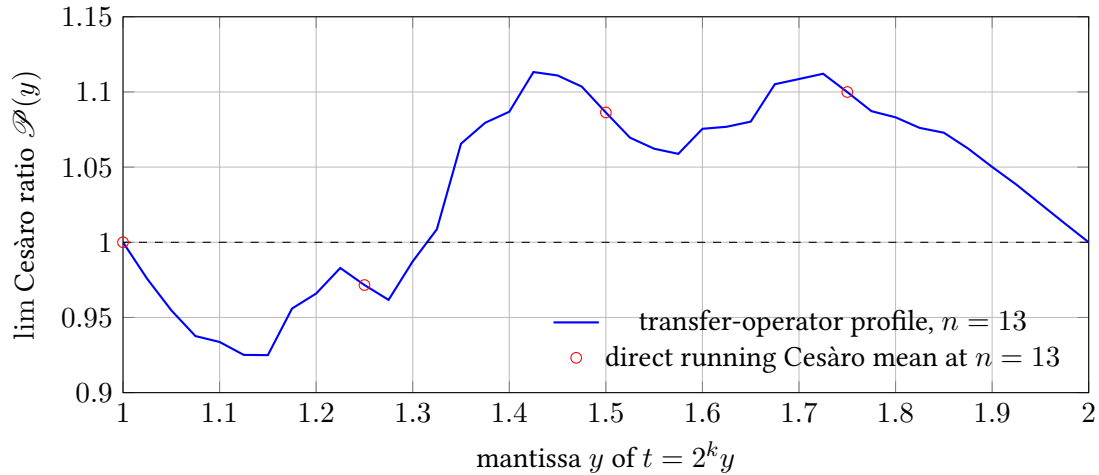


Figure 1: The dyadic-mantissa profile obstructing (Q2): the limit of the Cesàro ratio at $t = 2^k y$ as a function of y , for the normalized gauge $A_1 g_1$. The curve is the transfer-operator prediction (partial P_{13} -mass distribution); the circles are wholly independent running Cesàro means over $[8, 2^{13}y]$ computed by direct quadrature of $|f|$. The sharp mass accretion around $y = 4/3$ reflects the concentration of the eigenmeasure near the preimages of the Gelfond-maximizing cycle $\{1/3, 2/3\}$.

6 Interpretation in detail, I: the envelope of the lobe maxima

The zeros of f at the integers divide $(0, \infty)$ into unit lobes, and $|f|$ has exactly one maximum

$$E_n = \max_{n \leq x \leq n+1} |f(x)| = |f(\xi_n)|, \quad \xi_n \in (n, n+1),$$

per lobe, with signed value $(-1)^{s_2(n)} E_n$ (audited, and both halves formalized — `existsUnique_i sMaxOn_lobe` and `rvachevFourierProduct_eq_thueMorse_sign_mul_norm`, see section 2; the sign law is also the Atlas’s “sign of the Fabius sinc product”). Reading the question as “find the

envelope of the sequence E_n is natural — it is literally the subject of the companion question [7] — and this section shows exactly how far that reading can be pushed. The outcome is structural: *the peak sequence reproduces the whole decay spectrum of section 3 in miniature*, so it has a sharp limsup envelope, a zero liminf on that envelope's scale, a Cesàro envelope with the same exponent κ_1 as $|f|$ itself, and a density-one law with the typical exponent κ_0 — but no single envelope.

6.1 Geometry of one lobe, and a peak–integral comparison

Two audited facts anchor everything: away from the integers

$$(\log |f|)'(x) = -2x \sum_{k \geq 1} \frac{1 + v_2(k)}{k^2 - x^2}, \quad (\log |f|)''(x) = -2 \sum_{k \geq 1} (1 + v_2(k)) \frac{k^2 + x^2}{(k^2 - x^2)^2} < 0, \quad (7)$$

so every lobe of $|f|$ is strictly log-concave with a unique peak. That conclusion is formalized, by a route that does not pass through (7): `LobeLogFactorization` proves the lattice log-series identity $\log \|\Phi(x)\| = \sum_{(h,r)} \log(1 - x^2/(2^h(r+1))^2)$ at every non-lattice real x , `CentralLobeConcavity` and `SideLobeConcavity` give strict negativity of its termwise second derivative on the central and on every side lobe, and `OnePeakPerLobe` assembles them. The closed derivative formulas (7) themselves are not separately stated there. The following comparison is the workhorse of the mean layer; it does not appear in any of the four documents, and it is not formalized either — the log-concavity input is, the quantitative $\delta_{\min}$ and curvature bookkeeping is not.

Lemma 6.1 (peak–integral comparison). *There is an absolute constant C such that for all $n \geq 2$,*

$$\int_n^{n+1} |f| \leq E_n \leq C(1 + \log n)^3 \int_n^{n+1} |f|.$$

Proof. The left inequality is trivial. For the right one, write $d_0 = 1 + v_2(n)$, $d_1 = 1 + v_2(n+1)$, $\delta_0 = \xi_n - n$, $\delta_1 = n+1 - \xi_n$, and note $d_0, d_1 \leq 1 + \log_2(n+1)$. Splitting off the $k = n$ and $k = n+1$ terms of (7) and bounding the rest crudely ($|k^2 - x^2| \geq |k - x|n$ for $k \leq 2n$, quadratic tail beyond, $1 + v_2(k) \leq 1 + \log_2 k$) gives, for x in the middle segment $[n + \delta_0/2, n+1 - \delta_1/2]$,

$$(\log |f|)'(x) = \frac{d_0}{x-n} \theta_0(x) - \frac{d_1}{n+1-x} \theta_1(x) + \mathcal{O}(\log^2 n), \quad \theta_i(x) \in [\frac{9}{10}, \frac{6}{5}],$$

and

$$|(\log |f|)''(x)| \leq \frac{3d_0}{(x-n)^2} + \frac{3d_1}{(n+1-x)^2} + C \log n.$$

At the peak the derivative vanishes; since one of δ_0, δ_1 is at least $\frac{1}{2}$, the vanishing forces the other to satisfy $d_i/\delta_i \leq C \log^2 n$, whence $\delta_{\min} \geq c/\log^2 n$ and, on the middle segment around ξ_n , the curvature is at most $\Lambda \leq C \log^5 n$. By log-concavity, $|f(\xi_n \pm s)| \geq E_n e^{-\Lambda s^2/2} \geq E_n/e$ for $s \leq \min(\Lambda^{-1/2}, \delta_{\min}/2) \geq c(1 + \log n)^{-5/2}$, and integrating over that subinterval gives $\int_n^{n+1} |f| \geq c E_n (1 + \log n)^{-5/2}$, which is stronger than claimed. $\square$

Numerically the comparison is far tamer than the worst-case bound: over the shells $k = 9$ –13 the per-lobe ratio $E_n / \int_n^{n+1} |f|$ has median 1.992, and its maximum (4.55 at $k = 9$, 5.93 at $k = 13$, growing roughly linearly in k) is attained each time by the lobe *abutting* 2^{k+1} , i.e. the lobe pressed against the highest-multiplicity zero of the shell — the same two-adic geometry as the valleys.

6.2 The peak sequence is the spectrum in miniature

Theorem 6.2 (four layers of the peak sequence). *Let $\mathcal{E}(x) = \exp(-(\log x)^2/(2 \log 2))$.*

1. Sharp upper envelope. $\limsup_{n \rightarrow \infty} (\log E_n + \frac{(\log n)^2}{2 \log 2}) / \log n = -\kappa_\infty$, and no larger exponent works: the limsup is attained along the lobes containing $2^k \cdot \frac{4}{3}$ (equivalently $2^k \cdot \frac{2}{3}$). At second order the annular maxima carry the Lambert-W correction of Documents 1–2, beginning $-(\log \log x)^2 / (2 \log 2)$.
2. Zero liminf on that scale. $\liminf E_n / (\mathcal{E}(n)n^{-\kappa_\infty}) = 0$: along $n = 2^k$ (and $n = m2^k$, odd m) the peaks obey the two-adic law $E_{m2^k} \sim \frac{H(1)|H(m)|}{e^{(k+1)}} (m2^k)^{-(k+1)}$, whose leading quadratic coefficient $-1/\log 2$ is twice the universal one. Within the shell $[2^k, 2^{k+1})$ the deepest lobe observed is the one abutting 2^{k+1} (zero multiplicity $k+2$; confirmed at $k \in \{9, 11, 13\}$ by computing every lobe of the shell), with the lobe after 2^k (multiplicity $k+1$) its two-adic twin at the left edge; the two-adic law itself is verified through $k = 16$ (section 10.5).
3. Mean layer: exponent κ_1 again. By lemma 6.1,

$$\int_{2^k}^{2^{k+1}} |f| \leq \sum_{2^k \leq n < 2^{k+1}} E_n \leq C k^3 \int_{2^k}^{2^{k+1}} |f|,$$

so the shell means of the peak sequence have the same power $\kappa_1 = 2.748070 \dots$ as the shell means of $|f|$, and every Cesàro-type statement of theorem 5.1 transfers to the peak sequence up to the polylogarithmic slack. Numerically the slack is a sharp constant: over $k = 9, \dots, 14$,

$$\frac{\sum_{2^k \leq n < 2^{k+1}} E_n}{\int_{2^k}^{2^{k+1}} |f| dx} = 1.9238, 1.9237, 1.9237, 1.9237, 1.9237, 1.9238 (\pm 3 \cdot 10^{-5}),$$

supporting the conjecture that the ratio converges to a constant $\Theta_{\text{peak}} = 1.92372(3)$; granting it, the peak sequence has the Cesàro gauge $A_1^{\text{peak}} x^{-\kappa_1} \mathcal{E}(x)$ with $A_1^{\text{peak}} = \Theta_{\text{peak}} A_1 = 0.175569 \dots$

4. Typical layer: exponent κ_0 . For every $\varepsilon > 0$,

$$\frac{1}{2^k} \# \left\{ 2^k \leq n < 2^{k+1} : \left| \log E_n + \frac{(\log n)^2}{2 \log 2} + \kappa_0 \log n \right| \leq \varepsilon k \right\} \longrightarrow 1.$$

Proof. (i) The upper bound is the uniform envelope (audited); sharpness is the exact ray $y = 4/3$. (ii) is Documents 2 and 4's theorem (audited; verified numerically with the slow $1/k$ -type approach documented in section 10.5). (iii) is immediate from lemma 6.1 after summing over the shell. (iv) requires an argument, given now.

Lower half (few lobes are too small). Write $x = 2^k y$, $z = y - 1$, so that by the factorization and the κ -algebra,

$$\log |f(x)| + \frac{(\log x)^2}{2 \log 2} + \kappa_0 \log x = \Sigma_{k+1}(z) + \Psi(y), \quad \Sigma_m(z) := \sum_{j=0}^{m-1} \psi(T^j z),$$

with $\psi = \log |2 \sin \pi \cdot|$ centered and Ψ bounded on $[1, 2)$. By the Birkhoff theorem $\Sigma_m/m \rightarrow 0$ a.e., hence in measure: $\text{Leb}\{z : |\Sigma_{k+1}(z)| > \frac{\varepsilon}{2} k\} \rightarrow 0$. A lobe occupies a z -cell of length 2^{-k} ; if the cell contains a point of the good set, then $E_n \geq |f|$ there exceeds the κ_0 -line by more than $-\varepsilon k$. The number of cells disjoint from the good set is at most $2^k \cdot \text{Leb}(\text{bad}) = o(2^k)$.

Upper half (few lobes are too large). Call a cell J regular if for every $j \leq k' := k - 4 \lceil \log_2 k \rceil$ the image $2^j J$ keeps distance at least $k^2 2^{j-k}$ from $\mathbb{Z}$. For each j the offending set is 2^j intervals of length $2k^2 2^{-k}$ each, so at most $(2k^2 + 2)2^j$ cells are irregular at level j , and summing over $j \leq k'$ gives at most $C2^k/k^2 = o(2^k)$ irregular cells. On a regular cell every retained factor has logarithmic oscillation at most π/k^2 across the cell (length times $\pi 2^j$ cot-bound), so

$$\max_J P_{k+1} \leq e^{\pi/k} P_{k'}(z_J) \quad \text{for every } z_J \in J,$$

after discarding the last $4\lceil \log_2 k \rceil$ factors, each bounded by one. Now fix ε and choose $q \in (0, 1)$ with $\log \varrho_q < -q(1 - \frac{\varepsilon}{2}) \log 2$, possible because $(\log \varrho_q)/(q \log 2) \rightarrow -1$ as $q \downarrow 0$ (the pressure slope at the Lebesgue equilibrium; with the RPF package for $\mathcal{L}_q$ on bounded-variation functions — the weights s^q have total variation 2 uniformly in $q \in [0, 1]$ — this is standard perturbation theory, and it is the one imported input here, the same package as [section 5.2](#)). Markov's inequality with $I_q(k') \leq C_q \varrho_q^{k'}$ then bounds the measure of $\{P_{k'} \geq e^{-(1-\varepsilon)k \log 2}\}$ by $e^{-c(\varepsilon)k}$; since on a regular cell all points give comparable $P_{k'}$, the number of regular cells whose maximum exceeds the threshold is $2^k e^{-c(\varepsilon)k} = o(2^k)$. For such good cells, $E_n \leq C \cdot \text{shell factor} \cdot e^{-(1-\varepsilon)k \log 2}$ lies below the κ_0 -line by at most $\varepsilon k \log 2 + \mathcal{O}(\log k)$. Adjusting ε absorbs the constants. $\square$

Proposition 6.3 (central limit law for the peaks). *For every $u \in \mathbb{R}$,*

$$\frac{1}{2^k} \# \left\{ 2^k \leq n < 2^{k+1} : \log E_n + \frac{(\log n)^2}{2 \log 2} + \kappa_0 \log n \leq u \frac{\pi}{2} \sqrt{k} \right\} \rightarrow \Phi(u),$$

where Φ is the standard normal distribution function.

Proof. All ingredients are already on the table. The continuous CLT $\Sigma_k/\sqrt{k} \Rightarrow \mathcal{N}(0, \pi^2/4)$ holds by Document 1's martingale argument with the variance of [theorem 9.1](#). Convergence in distribution under Lebesgue measure transfers to the cell ensemble because, outside the $o(2^k)$ irregular cells, all points of a cell give the same $\Sigma_{k'}$ up to π/k , and the discarded top factors change $\log E_n$ by at most $\mathcal{O}(\log^2 k)$ in one direction and, in the other, by at least $\log$ of the product of the discarded factors on a nested subcell: choosing inside the cell, factor by factor from the finest down, the third of the current subcell on which the new factor exceeds $\frac{1}{2}$ leaves a nonempty subcell on which the discarded product is at least $2^{-4\lceil \log_2 k \rceil}$. Both corrections are $\mathcal{O}(\log^2 k) = o(\sqrt{k})$, as is the bounded mantissa term, so the cell statistics of $\log E_n$ inherit the Gaussian limit with the same normalization. $\square$

The measured quantiles agree on the left and are compressed on the right, exactly as they must be at finite k : standardized quantiles at $k = 13$ are $(-1.63, -0.43, +0.05, +0.36, +0.65)$ at levels $(5, 25, 50, 75, 95)\%$ against the Gaussian $(-1.645, -0.674, 0, +0.674, +1.645)$. The median sits on the κ_0 -line to within 0.05 standard units — the typical exponent is visibly κ_0 — while the right tail is truncated by the deterministic Gelfond cap: no peak can exceed the κ_∞ -line, which sits only $(\kappa_0 - \kappa_\infty) \log 2 \cdot \sqrt{k}/(\pi/2) \approx 0.35\sqrt{k}$ standard units above the center (≈ 1.26 at $k = 13$). The truncation recedes like $\sqrt{k}$, so the Gaussian limit opens up only logarithmically slowly in x — the same finite-size phenomenon that hides the LIL constant from small- L simulations in [section 9](#).

6.3 The discrete dictionary: peaks as Thue–Morse spectra

Across the shell $[2^k, 2^{k+1})$, the lobe of $n = 2^k + m$ occupies the z -cell of $m2^{-k}$, and up to the bounded factors above, E_n is the windowed maximum of P_{k+1} there — that is, of the modulus of the Thue–Morse polynomial $\sum_{r < 2^{k+1}} (-1)^{s_2(r)} e^{2\pi i r z}$ near the dyadic frequency $m/2^k$. The peak-envelope interpretation is thus exactly the question of the value distribution of Thue–Morse spectra, and the exact grid identities of that world transfer:

Proposition 6.4 (two exact grid identities). *For every $n \geq 1$:*

$$(i) \quad \prod_{r=1}^{2^n-1} 2 \sin \frac{\pi r}{2^n} = 2^n, \quad (ii) \quad \sum_{m=0}^{2^n-1} P_n\left(\frac{m}{2^n}\right)^2 = 1.$$

Proof. (i) is the classical evaluation $\prod_{r=1}^{N-1} (1 - \omega^r) = N$ for $\omega = e^{2\pi i/N}$, taken in modulus. It says that the mean of $\psi = \log |2 \sin \pi \cdot|$ over the *nonzero* points of the 2^n -grid is $2^{-n} \log 2^n \rightarrow 0$ — the grid analogue of $\int_0^1 \psi = 0$, which is what makes the discrete Birkhoff sums behave like the continuous ones. (ii) Let $p(t) = \prod_{j < n} (1 - e^{2\pi i 2^j t})$, of degree $2^n - 1$ with coefficients ± 1 , so $|p| = 2^n P_n$. The 2^n -point

discrete Parseval identity gives $2^{-n} \sum_m |p(m/2^n)|^2 = \sum_r |c_r|^2 = 2^n$, i.e. $\sum_m P_n(m/2^n)^2 = 1$. (Verified to 10^{-12} at $n = 6, 10, 14$; even m contribute exactly zero.) $\square$

Both identities are formalized, each in a more general form. (i) is `SineProductRootsUnity`'s `prod_two_sin_pi_div_two_pow`, a specialization of `prod_two_sin_pi_div`, which proves $\prod_{r=1}^{N-1} 2 \sin(\pi r/N) = N$ for every $N \geq 1$. (ii) is `DiscreteLacunaryParseval`'s `sum_prod_sin_sq_two_pow`, a specialization of the master discrete orthogonality theorem `sum_cos_mul_rieszProduct`, which holds for arbitrary Hadamard-gap integer frequencies, amplitudes and phases on any grid of size $N \geq m_n$ — the dyadic case $N = 2^n, m_j = 2^j$ sitting at the edge of that admissible resolution. Both Lean statements are for every $n \geq 0$.

Identity (ii) is the discrete twin of $\int_0^1 P_n^2 = 2^{-n}$: the $q = 2$ layer is rigid in both the continuous and the sampled world, a point taken up in [section 7](#).

6.4 Verdict for this interpretation

No function G satisfies $E_n/G(n) \rightarrow c > 0$: the four layers of [theorem 6.2](#) are separated by genuine powers of x , and even within one layer the fluctuations are of iterated-logarithm size. What the interpretation *does* admit is exactly what (Q2)–(Q3) admit for $|f|$: a bounded Cesàro normalization with the same exponent κ_1 (via [lemma 6.1](#)), with the empirical constant $\Theta_{\text{peak}} = 1.9237$ tying the two readings together. For the companion question [7], the sharp answers are the limsup envelope $x^{-\kappa_\infty} \mathcal{E}(x)$ and the typical law $x^{-\kappa_0 + o(1)} \mathcal{E}(x)$ — two different formulas, necessarily.

7 Interpretation in detail, II: the root-mean-square amplitude

The RMS reading asks for g such that the quadratic Cesàro mean of $(f/g)^2$ is tame. Because $f = \widehat{\text{up}}$, the shell RMS of f is the energy spectral envelope of the up-function, and this is the one reading in which the deleted draft's exponent $\log_2(2\pi)$ is correct. It is also, by a wide margin, the most rigid layer of the whole problem: its per-level mean is exact, its subleading spectrum is computable in closed form, and its envelope constant converges at a clean geometric rate with alternating sign.

7.1 Exactness at every scale

The continuous identity $\int_0^1 P_n^2 = 2^{-n}$ ([proposition 5.4](#)) and the discrete identity $\sum_m P_n(m/2^n)^2 = 1$ ([proposition 6.4](#)) leave no asymptotics in the $q = 2$ numerator: both the Lebesgue mean and the grid mean of P_n^2 equal 2^{-n} exactly, for every n . All remaining structure sits in the bounded mantissa weight, and it too is spectrally transparent:

Proposition 7.1 (exact subleading spectrum of $\mathcal{L}_2$). *The operator $(\mathcal{L}_2 h)(x) = \frac{1}{2} [\sin^2(\frac{\pi x}{2}) h(\frac{x}{2}) + \cos^2(\frac{\pi x}{2}) h(\frac{x+1}{2})]$ satisfies, exactly,*

$$\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \mathbf{1}, \quad \mathcal{L}_2 [\sin 2\pi x] = -\frac{1}{4} \sin 2\pi x, \quad \mathcal{L}_2 [1 + 3 \cos 2\pi x] = -\frac{1}{4} (1 + 3 \cos 2\pi x),$$

and on even Fourier modes it acts by frequency halving, $\mathcal{L}_2 e^{2\pi i 2mx} \mapsto \frac{1}{2} e^{2\pi i mx}$.

Proof. With $e_m(x) = e^{2\pi i mx}$: for even frequency $2m$, $e^{2\pi i 2m \frac{x}{2}} = e_m(x)$ on both branches and $\sin^2 + \cos^2 = 1$ gives $\mathcal{L}_2 e_{2m} = \frac{1}{2} e_m$ (the $m = 0$ case is the first display). For $m = 1$,

$$\mathcal{L}_2 e_1 = \frac{1}{2} e^{\pi i x} [\sin^2(\frac{\pi x}{2}) - \cos^2(\frac{\pi x}{2})] = -\frac{1}{2} e^{\pi i x} \cos \pi x = -\frac{1}{4} (e_1 + \mathbf{1}),$$

and likewise $\mathcal{L}_2 e_{-1} = -\frac{1}{4} (e_{-1} + \mathbf{1})$. Hence $\{\mathbf{1}, e_1, e_{-1}\}$ spans an invariant subspace on which $\mathcal{L}_2$ has matrix eigenvalues $\{\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4}\}$; the real eigenfunctions for $-\frac{1}{4}$ are $\sin 2\pi x = (e_1 - e_{-1})/2i$ and $1 + 3 \cos 2\pi x$ (direct check: $\mathcal{L}_2 [3 \cos 2\pi x] = -\frac{3}{4} (1 + \cos 2\pi x)$, so $\mathcal{L}_2 [1 + 3 \cos] = \frac{1}{2} - \frac{3}{4} - \frac{3}{4} \cos = -\frac{1}{4} (1 + 3 \cos)$). $\square$

All four displays of [proposition 7.1](#) are formalized in `RMSTransferEigenfunctions`, in exactly this normalization of $\mathcal{L}_2$: `rms_transfer_const_eigen`, `rms_transfer_sin_eigen`, `rms_transfer_one_add_cos_eigen`, and the mode halving in its real form, `rms_transfer_cos_even_mode` together with `rms_transfer_sin_even_mode`. They are stated pointwise in x — elementary trigonometry, no operator object and no spectral theory — so what is formal is the *eigen-identities*, not the claim that $-\frac{1}{4}$ is the subleading eigenvalue, which rests on the collocation bound below.

Collocation places the rest of the spectrum below $3 \cdot 10^{-4}$ in modulus, so the shell-RMS ratios converge at the exact geometric rate $\lambda_2/\lambda_1 = -\frac{1}{2}$ per shell — relative error $\mathcal{O}(1/x)$, with alternating sign. This is visible to the eye in the measurements: the shell means of $(f/g_{\kappa_2})^2$ for $k = 4, 5, 6, 7, 8$ differ from their limit by $+1.51 \cdot 10^{-5}$, $-7.6 \cdot 10^{-6}$, $+3.8 \cdot 10^{-6}$, $-1.9 \cdot 10^{-6}$, $+0.94 \cdot 10^{-6}$ — each step multiplying the error by $-\frac{1}{2}$ to the displayed accuracy — with limit

$$A_2^2 = 0.0106108372\dots, \quad A_2 = 0.103008918\dots$$

(direct quadrature and the spectral iteration agree; the L^1 layer, by contrast, has the non-elementary subleading pair $-0.13435 \pm 0.0500i$ of modulus 0.1434, rate $|\lambda_2|/\varrho_1 = 0.217$ per shell).

7.2 The RMS answer to the averaged question

Theorem 7.2 (RMS version of [theorem 5.1](#)). *Let $G_2 = A_2 x^{-\kappa_2} \mathcal{E}(x)$ with $\kappa_2 = \log_2(2\pi)$. Then: (i) the quadratic Cesàro ratio $\frac{1}{t-t_0} \int_{t_0}^t (f/G_2)^2$ is eventually confined to a compact subinterval of $(0, \infty)$ and tends to 1 along $t = 2^k$ with relative error $\mathcal{O}(2^{-k})$; (ii) for $t = 2^k y$ with fixed $y \in (1, 2)$ it converges to a continuous, nonconstant mantissa profile $\mathcal{P}_2(y)$, so the limit for all t fails — for the same reason as at $q = 1$: the $\mathcal{L}_2$ -eigenmeasure ν_2 is atomless and singular; (iii) the logarithmic mean $\frac{1}{\log(t/t_0)} \int_{t_0}^t (f/G_2)^2 \frac{dx}{x}$ converges for all $t \rightarrow \infty$.*

Proof. (i) and (iii) are verbatim the proofs of [theorem 5.1](#)(i)–(ii) and [theorem 5.7](#) with (P_n, ϱ_1, W) replaced by $(P_n^2, \frac{1}{2}, W_2)$; the improved rate in (i) is [proposition 7.1](#). For (ii), only the two measure statements change. *Atomless*: the eigenmeasure equation gives $\nu_2(\{Tz\}) = \nu_2(\{z\})/\sin^2(\pi z) \geq \nu_2(\{z\})$, with equality only at $z = \frac{1}{2}$. No forward orbit can linger where $\sin^2$ is near 1: if $w \in [\frac{3}{8}, \frac{5}{8}]$ then $Tw \in [0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$, where $\sin^2 \leq \frac{1}{2}$; and off $[\frac{3}{8}, \frac{5}{8}]$ one has $\sin^2 \leq \sin^2(\frac{3\pi}{8}) < 0.854$. So along any orbit the atom mass is nondecreasing and gains a factor $\geq 1/0.854$ at least every second step, forcing it to zero whether the orbit is eventually periodic (mass grows around each loop) or not (infinitely many distinct atoms of nonvanishing mass). *Singular*: an absolutely continuous part would have density r with $\sin^2(\pi z) r(Tz) = \frac{1}{2}r(z)$, hence $r(T^n z) = 2^{-n}r(z)/P_n(z)^2$; for a.e. z the Birkhoff theorem gives $P_n(z)^2 = e^{-2n \log 2(1+o(1))}$, so on $\{r > 0\}$ the density grows like $e^{n(\log 2 + o(1))}$, contradicting $r \in L^1$ by the same Borel–Cantelli argument as before. (The general criterion behind both cases: the q -eigenmeasure is singular whenever $\varrho_q e^{q \log 2} > 1$, i.e. whenever $\kappa_q < \kappa_0$; at $q = 2$ this reads $\frac{1}{2} \cdot 4 = 2 > 1$.) The nonconstancy of $\mathcal{P}_2$ then follows as in [theorem 5.5](#). $\square$

The measured L^2 profile (same transfer computation as [figure 1](#), weight $W_2 P_{13}^2$) spans $[0.894, 1.192]$ — wider than the L^1 profile $[0.925, 1.113]$, as expected, since squaring weights the singular concentration near the Gelfond preimages more heavily. So even the most rigid reading of the question keeps the dyadic phase obstruction; what it gains is exact constants and an $\mathcal{O}(1/x)$ rate everywhere the limit does exist.

7.3 Plancherel, Poisson, and the total energy

Proposition 7.3 (total energy).

$$\int_{\mathbb{R}} f(x)^2 dx = \int_{\mathbb{R}} \text{up}(t)^2 dt = \frac{1}{2} + \sum_{m=0}^{\infty} f\left(m + \frac{1}{2}\right)^2 = 0.808873535967972\dots$$

Proof. The first equality is Plancherel ($f = \widehat{\text{up}}$, real and even). For the second, expand up, supported in $[-1, 1]$, in Fourier series on $[-1, 1]$: the coefficients are $\frac{1}{2}\widehat{\text{up}}(k/2) = \frac{1}{2}f(k/2)$, and Parseval gives $\int \text{up}^2 = \frac{1}{2} \sum_{k \in \mathbb{Z}} f(k/2)^2$; the even $k \neq 0$ terms vanish (f vanishes at nonzero integers), $k = 0$ gives $\frac{1}{2}$, and the odd terms pair up. Numerically the direct quadrature $2 \int_0^{2^{14}} f^2$ and the series agree to $2 \cdot 10^{-15}$, the displayed value; the series converges with super-Gaussian speed since $f(m + \frac{1}{2})^2$ decays like $\mathcal{E}(m)^2$. $\square$

This is a satisfying exactness check on the entire numerical apparatus, and a natural first formalization target on the L^2 side: the ingredients (compact support, the sinc product, Parseval on an interval) are all present in the Lean development (`FourierProduct`, `PoissonSummation`).

7.4 The lognormal fallacy, quantified

Why did plausible heuristics — the deleted draft included — land on wrong exponents? Because every mean-type interpretation lives in the large-deviation regime of the lacunary product, where the correct generating function is the pressure $q \mapsto \log \varrho_q$, not the CLT parabola. The exact offset formula (immediate from the definitions) is

$$\kappa_0 - \kappa_q = 1 + \frac{\log \varrho_q}{q \log 2},$$

giving the exact value $\frac{1}{2}$ at $q = 2$; a lognormal extrapolation of the CLT would instead predict $\kappa_0 - \kappa_q^{\text{LN}} = q\sigma^2/(2 \log 2)$ with $\sigma^2 = \pi^2/4$. The comparison:

q	ϱ_q	κ_q (true)	κ_q^{LN} (lognormal)
0.25	0.872813205188	2.936517	2.706533
0.5	0.786052008490	2.846103	2.261569
1	0.661322602061	2.748070	1.371642
2	0.5 (exact)	2.651496	−0.408211
4	0.320194101601	2.562241	−3.967919
∞	—	2.359015	$-\infty$

Already at $q = 2$ the lognormal prediction is not merely off — it is negative, i.e. it predicts moment *growth* relative to the typical scale that the function cannot deliver: the deterministic Gelfond cap $P_n \leq C(\sqrt{3}/2)^n$ truncates the upper tail that a lognormal law would need. The pressure slope $(\log \varrho_q)/(q \log 2)$, measured as $-0.785, -0.695, -0.597$ at $q = \frac{1}{4}, \frac{1}{2}, 1$, bends away from its $q \rightarrow 0$ limit -1 immediately: the moments leave the Gaussian bulk at once. Three consequences deserve record. First, the draft’s exponent is exactly the $q = 2$ row — a correct constant attached to the wrong norm. Second, the draft’s proposed $(\log x)^{-1/2}$ factor is refuted most decisively in this, its most favorable, reading: the shell RMS normalization converges to the *constant* A_2 at a geometric rate, so no logarithmic factor of any exponent survives. Third, the spacing of the four headline exponents $\kappa_\infty < \kappa_2 < \kappa_1 < \kappa_0$ is itself the multifractal signature of the dyadic dynamics; treating any one of them as “the” decay rate discards the others’ regimes, which is precisely the category error the original conjecture made.

8 Audit of the four documents

The four documents pairwise overlap heavily and agree on every overlapping claim except one (the iterated-logarithm constant, [section 9](#)). Documents 1 and 2 analyze the pointwise, typical, and extremal regimes; Documents 3 and 4 additionally analyze the mean regimes and are the only two that answer the question actually asked.

8.1 Document 1 (*Decay of the Fourier Transform ...: Dyadic products, ergodic fluctuations, and sharp annular envelopes*)

Content. Exact dyadic decomposition; zero multiplicities and Thue–Morse signs; one extremum per unit interval; typical-ray law with $\kappa_{\text{typ}} = \kappa_0$; a central limit theorem with per-step variance $\pi^2/4$ via the martingale difference $d = \log |\tan \pi x|$; a law of the iterated logarithm with constant $(\pi/2)\sqrt{2N \log \log N}$; the maximal invariant average $\beta = \log(\sqrt{3}/2)$ with the slow-ray exponent $\kappa_* = \kappa_\infty$; and the annular maximum $\log M_N = -\frac{a}{2}N^2 + (\beta - \log \pi - \frac{a}{2})N - \frac{(\log N)^2}{2a} + \frac{\log N \log \log N}{a} + \mathcal{O}(\log N)$.

Verified. The exact decomposition (section 4); the algebra of its centered form (resymbolized independently); κ_0, κ_* to all printed digits; the variance $\pi^2/4$ and covariances $\pi^2/(12 \cdot 2^k)$ (Monte Carlo and quadrature, section 10.6); $\int_0^1 \log^2 |\tan \pi x| dx = \pi^2/4$; $Pd = 0$; the Green–Kubo resummation $\pi^2/12 + 2 \sum_k \pi^2/(12 \cdot 2^k) = \pi^2/4$ in its appendix; the annular expansion’s first three terms agree with Document 2’s Lambert form after expanding W .

Defects. (i) **Scope:** the document never addresses the Cesàro question (Q2)–(Q3); its conclusions answer the deleted draft’s pointwise claim instead of the Stack Exchange question. Within the question’s actual reading it is a (high-quality) analysis of the wrong functional. (ii) The boundary-layer theorem’s error term $\mathcal{O}(\log N)$ is coarser than Document 2’s $\mathcal{O}((\log \log N)^2)$; not an error, but strictly subsumed. (iii) No numerical validation is supplied; all of its constants nevertheless survived this audit’s numerics. Notably, its fluctuation section is the one place in the entire corpus that contradicts—and, as section 9 shows, corrects—the published literature.

8.2 Document 2 (*Sharp Real-Axis Decay ...: Lambert W boundary layers, and two-adic troughs*)

Content. Everything in Document 1’s extremal track, sharpened: the annular maximum in closed Lambert form,

$$\log M_k = -\frac{\lambda}{2}k(k+1) - (k+1) \log \pi + (k+1)\beta - \frac{W(c(k+1))^2 + 2W(c(k+1))}{2\lambda} + \mathcal{O}((\log \log k)^2), \quad c = \frac{\lambda}{\pi} \sqrt{\frac{3}{2}},$$

an elementary sharp Gelfond bound $Q_n \leq \sqrt{5/3} (\sqrt{3}/2)^n$ via an explicit algebraic subaction, two-adic trough asymptotics $E_{m2^k} \sim \frac{H(1)|H(m)|}{e^{(k+1)}} (m2^k)^{-(k+1)}$ for odd m , a numerical table of $\log M_k$, and—critically—an explicit section stating that the Cesàro question is a *different*, L^1 , transfer-operator problem which the document does not solve.

Verified. The subaction identity $3V(u) - 4uV(4u(1-u)) = \frac{(2u+1)(4u-3)^2}{3}$, $V(u) = 1 + \frac{2}{3}u$ (expanded symbolically: exact); the annular table: this audit’s independent maximization reproduces all ten rows of its $(\log M_k, x^*/2^k)$ table to every printed digit (10^{-6}), $k = 3, \dots, 12$ (section 10.4); $H(1) = 0.55377127588881009534\dots$ (recomputed to 30 digits, matches all 20 printed); the Lambert residuals (bounded, -0.72 to -0.82 , consistent with the claimed $\mathcal{O}((\log \log k)^2)$ window); the trough law for $m = 1$ and $m = 3$ (section 10.5); the predicted drift of the annular maximizer from mantissa $7/6$ toward deeper preimages of $1/3$ —observed exactly at $k = 14$, where the maximizer jumps from $x^*/2^k \approx 1.16667$ to $\approx 1.08333 = 13/12$.

Defects. (i) **Scope:** same as Document 1, but self-aware — its §9 correctly scopes the Cesàro problem to the transfer-operator framework that Documents 3 and 4 then execute; among the four only Document 2 draws this boundary explicitly. (ii) The two-adic trough asymptotic converges slowly; at $k = 16$ the measured $\log E_{2^k}$ still exceeds the limit formula by 0.098 (a 10% multiplicative residual, decaying roughly like $1/k$). The statement is asymptotically consistent but numerically “young” at accessible k ; a one-line remark on the convergence speed would have prevented over-reading. (iii) No fluctuation theory (deliberate).

8.3 Document 3 (*Sharp Decay Laws ...: extremal envelopes, metric fluctuations, and Cesàro normalization*)

Content. The widest scope: exact factorization; typical law; one-sided iterated-logarithm fluctuations (quoted from [1]); extremal envelope with an elegant ergodic-optimization proof (pushforward to the logistic map, inequality $z^3(1-z) \leq 27/256$); the L^1 theory: Fouvry–Mauduit asymptotic, transfer operator (3), $\lambda = 0.661322602060565$, $\beta_1 = \kappa_1$, an explicit gauge, the bounded-Cesàro theorem, the dyadic-endpoint limit, and the singular-eigenmeasure obstruction to (Q2); the exact L^2 identity and the identification of the deleted draft’s exponent as κ_2 ; the L^p pressure spectrum with endpoints $p \downarrow 0$, $p \rightarrow \infty$; Chebyshev-collocation numerics.

Verified. The ergodic-optimization lemma (equality case included); λ to thirteen digits by two independent methods; κ_1 (the printed 2.748070014871334 against this audit’s 2.748070014871335 from $\varrho_1 = 0.661322602060565$: the discrepancy is one unit in the last place, inside the $\pm 5 \cdot 10^{-15}$ uncertainty inherited from ϱ_1 ’s own last digit); the collocation table (all displayed eigenvalues reproduced to 10^{-15}); κ_1 (its printed 2.748070014871334 is exactly the value of $\frac{1}{2} + \log_2(\pi/\varrho_1)$ at $\varrho_1 = 0.661322602060565$, recomputed here in 25-digit arithmetic); the bounded-Cesàro theorem and the dyadic-endpoint limit (measured, section 10.2); the singularity argument for the equilibrium measure (checked line by line; the inequality $\log \lambda > -\log 2$ that it needs follows rigorously from $\varrho_1 \geq 3^{-1/2}$, proposition 5.3); the Fouvry–Mauduit and Aistleitner–Hofer–Larcher attributions (verbatim against the published paper and the authors’ arXiv source); the L^2 identity; the identification $\frac{1}{2} + \log_2(\pi\sqrt{2}) = \log_2(2\pi)$.

Defects. (i) **Its Theorem 4.2 (sharp upper metric fluctuation) states** $\limsup_n S_n(t)/\sqrt{n \log \log n} = \pi$ **a.e.** The quotation is a *faithful* rendering of the published Theorem 1.2 of [1], but the published constant is itself erroneous by a factor $\sqrt{2}$; the correct value is $\pi/\sqrt{2}$ (section 9). Every restatement downstream (its (LIL- x) display and the abstract’s “excursion” constant) inherits the factor. The order $\sqrt{\log \xi \log \log \log \xi}$, and every conclusion drawn from it (no fixed power of $\log x$ can flatten the amplitude), are unaffected. (ii) Minor: the claim that its gauge has “eventually monotone derivatives of any order” is proved per order (thresholds X_m growing with m); this matches the question’s wording but is worth stating as such, since no common half-line works for all orders simultaneously.

8.4 Document 4 (*Decay ...: Dyadic renormalization, transfer-operator norms, sharp envelopes, and exceptional valleys*)

Content. The most systematic mean-theory: a master normalization identity; the full L^q shell spectrum with monotonicity; exact shell-geometric-mean constancy at $q = 0$; the RMS row by Parseval; the $q = 1$ row with a *rigorous, executable* Collatz–Wielandt/Sturm certificate for ϱ_1 ; the same two-part Cesàro conclusion as Document 3, but with complete proofs of the eigenmeasure’s atomlessness and singularity; a sharp two-step logarithmic inequality yielding the Gelfond bound; a dyadic-boundary identity giving valley asymptotics for every fixed offset r (with the zero-multiplicity $d = 1 + v_2(r+1)$ appearing as the power of $1/n$); a continuum of intermediate quadratic rates; and collocation numerics for $q \in \{1, 2, 3, 4, 8, 16\}$.

Verified. The master identity (numerically exact); the $q = 0$ exactness argument (each $j \geq 1$ level integrates to $-\log 2$ exactly on the shell); the two-step inequality (cubed form reduces to $\max u^3(1-u) = 27/256$: exact); the atomlessness and singularity proofs (sound; the Borel–Cantelli step uses only T -invariance of Lebesgue); $\varrho_3 = 0.394896632370234$, $\varrho_4 = 0.320194101601104$ (matching its table); **the certificate: executed verbatim under SymPy, all four Sturm root-counts return zero and the assertions pass, so $0.66126807 < \varrho_1 < 0.66134891$ holds with exact rational arithmetic**; the valley theorem for $r = 0$ (equivalent to Document 2’s $m = 1$ trough law; same slow convergence); the phase-profile formula $\mathcal{P}(y) = (1 + \mathcal{F}(2y-1))/(2y)$, realized numerically in figure 1.

Defects. (i) No fluctuation theory at all: between the a.e. law ($q = 0$ row) and the shell means

there is no CLT/LIL layer; Document 4 is silent exactly where Documents 1 and 3 collide. Since that collision is the corpus's one real error surface, the omission kept Document 4 clean but also unhelpful for adjudication. (ii) Its Perron–Frobenius proposition asserts a spectral gap on Hölder spaces for weights vanishing at the endpoints with a citation rather than a proof; standard, but it is the one analytic ingredient of its Cesàro theorem not reproved from scratch in any of the four documents. (iii) The “intermediate rates” section (quadratic coefficients $-(1 - \alpha + \alpha^2/2)/a$) is explicitly conditional on a typical-decay hypothesis along the moving offset; it is a heuristic bridge, correctly flagged as such, and was not counted as a theorem in this audit.

8.5 Relations among the documents

Document 1 $\subset$ Document 2 (extremal track, where 2 is sharper); Documents 3 $\approx$ 4 (mean track, where 4 is more rigorous on $q = 1$ and 3 is broader in scope). The intersection of all four is the exact factorization, κ_0 , κ_∞ , and the refutation of the deleted draft's pointwise claim. The union, minus Document 3's inherited LIL constant, plus the logarithmic-mean repair of [theorem 5.1\(iv\)](#), is – to the best of this audit's verification – a correct and essentially complete account of the problem.

9 The iterated-logarithm conflict and an error in the published literature

9.1 The conflict

For $S_L(t) = \sum_{j=0}^{L-1} \log |2 \sin(\pi 2^j t)|$:

- Document 1 proves a CLT with per-step variance $\pi^2/4$ (martingale differences $d = \log |\tan \pi x|$, $\int d^2 = \pi^2/4$, transfer-annihilation $Pd = 0$) and the LIL

$$\limsup_N \frac{S_N}{\sqrt{2N \log \log N}} = \frac{\pi}{2} \quad \text{a.e., i.e.} \quad \limsup_N \frac{S_N}{\sqrt{N \log \log N}} = \frac{\pi}{\sqrt{2}} = 2.2214 \dots$$

- Document 3 quotes the published Theorem 1.2 of [1]: for a.e. α , $\prod_{\ell=0}^L |2 \sin \pi 2^\ell \alpha| \leq \exp((\pi + \varepsilon)\sqrt{L \log \log L})$ eventually and $\geq \exp((\pi - \varepsilon)\sqrt{L \log \log L})$ infinitely often – constant $\pi = 3.1415 \dots$

These differ by exactly $\sqrt{2}$ and cannot both be sharp.

9.2 The rigorous adjudication

The conflict is decided by computing the variance exactly. Write $\psi(x) = \log |2 \sin \pi x|$, $S_m = \sum_{\ell=0}^{m-1} \psi \circ T^\ell$.

Theorem 9.1 (exact variance of the lacunary sums). *$\psi \in L^2(0, 1)$ with mean zero, and for every $m \geq 1$,*

$$\int_0^1 S_m(x)^2 dx = \frac{\pi^2}{4} m - \frac{\pi^2}{3} (1 - 2^{-m}). \quad (8)$$

In particular the functional (2.9) of [1] evaluates, for $f_1 = \psi$, to

$$\sigma_{f_1}^2 = \lim_{m \rightarrow \infty} \frac{1}{m} \int_0^1 S_m^2 = \frac{\pi^2}{4}, \quad \text{hence} \quad \sigma_{f_1} = \frac{\pi}{2}.$$

Proof. The Clausen expansion $\psi(x) = -\sum_{j \geq 1} \cos(2\pi j x)/j$ holds in $L^2(0, 1)$ (the coefficients are square-summable, and the series is the classical Fourier series of ψ). By Parseval, using

$$\int_0^1 \cos^2(2\pi jx) dx = \frac{1}{2},$$

$$c_0 := \int_0^1 \psi^2 = \frac{1}{2} \sum_{j \geq 1} \frac{1}{j^2} = \frac{\pi^2}{12},$$

and for $r \geq 1$, since $\int_0^1 \cos(2\pi jx) \cos(2\pi k2^r x) dx = \frac{1}{2} \mathbf{1}_{\{j=2^r k\}}$,

$$c_r := \int_0^1 \psi(x) \psi(2^r x \bmod 1) dx = \frac{1}{2} \sum_{k \geq 1} \frac{1}{2^r k} \cdot \frac{1}{k} = \frac{\pi^2}{12} 2^{-r}.$$

Stationarity of $(\psi \circ T^\ell)$ under Lebesgue measure gives $\int S_m^2 = mc_0 + 2 \sum_{r=1}^{m-1} (m-r)c_r$, and the elementary summation $\sum_{r=1}^{m-1} (m-r)2^{-r} = m - 2 + 2^{1-m}$ yields (8). (Check at $m = 1$: $\pi^2/4 - \pi^2/6 = \pi^2/12 = c_0$.) $\square$

Theorem 9.1 is formalized exactly as stated, and for every m including the empty sum $m = 0$: `BirkhoffVariance's integral_sq_birkhoff_sum_all` in Birkhoff-sum form, and `BirkhoffProductBridge's integral_sq_log_prod_two_sin_all` in the product form $\int_0^1 \log^2 |\prod_{k < m} 2 \sin(\pi 2^k t)|$. Its two inputs are formal as well: the covariance ladder $c_r = (\pi^2/12)2^{-r}$ for every lag r , over the genuine mod-one doubling map (`CovarianceLadder`, `integral_cocycle_covariance`), and both Clausen values, $\int_0^1 \log^2(2 \sin \pi t) dt = \pi^2/12$ (`CocycleParseval`, `integral_sq_log_two_sin_pi_mul`) and $\int_0^1 \log^2 |\tan \pi t| dt = \pi^2/4$ (`GordinParseval`, `integral_sq_log_abs_tan_pi_mul`). This is therefore the finite- m kernel of [finding 9.3](#) in machine-checked form; the limit theorems that consume it are not (see after [theorem 9.2](#)).

Theorem 9.2 (corrected sharp constant). *For Lebesgue-almost every α ,*

$$\limsup_{L \rightarrow \infty} \frac{S_L(\alpha)}{\sqrt{2L \log \log L}} = \frac{\pi}{2}, \quad \text{equivalently} \quad \limsup_{L \rightarrow \infty} \frac{\log \prod_{\ell=0}^L |2 \sin \pi 2^\ell \alpha|}{\sqrt{L \log \log L}} = \frac{\pi}{\sqrt{2}} = 2.2214 \dots$$

Proof (audit of Document 1's argument, with the boundary estimates completed). Set $d = 2\psi - \psi \circ T$. The double-angle identity gives $d(x) = \log |\tan \pi x|$, and $\int_0^1 d^2 = \frac{2}{\pi} \int_0^{\pi/2} \log^2 \tan u du = \frac{2}{\pi} \cdot \frac{\pi^3}{8} = \frac{\pi^2}{4}$. The transfer operator P of T (fiber average over the two preimages) annihilates d : $d(x/2) + d((x+1)/2) = \log |\tan \frac{\pi x}{2} \cdot \tan(\frac{\pi x}{2} + \frac{\pi}{2})| = \log 1 = 0$. Hence $(d \circ T^\ell)_{\ell \geq 0}$ is a stationary ergodic sequence of reverse martingale differences with respect to the decreasing filtration $(T^{-\ell} \mathcal{B})$, with orthogonal increments ($\int (P^k d) d = 0$ for $k \geq 1$) and variance $\pi^2/4$; note this matches [theorem 9.1](#), as it must, since telescoping gives the exact identity

$$S_L = \sum_{\ell=0}^{L-1} d \circ T^\ell - \psi + \psi \circ T^L.$$

The law of the iterated logarithm for stationary ergodic square-integrable (reverse) martingale difference sequences (Stout; Heyde–Scott; via Skorokhod embedding, whose time change has almost-surely linear growth $T_L/L \rightarrow \mathbb{E}d^2$ by the ergodic theorem) gives $\limsup (\sum_{\ell < L} d \circ T^\ell) / \sqrt{2L \log \log L} = \sqrt{\mathbb{E}d^2} = \pi/2$ a.e. The boundary terms are almost surely $o(\sqrt{L \log \log L})$: $\psi \leq \log 2$ above, while $\text{Leb}(\psi < -u) \leq \frac{2}{\pi} e^{-u}(1 + o(1))$ gives $\sum_L \mathbb{P}(\psi \circ T^L < -2 \log L) < \infty$, so by Borel–Cantelli $\psi(T^L \alpha) \geq -2 \log L$ eventually a.e. The product form follows by exponentiating and absorbing the index shift $L \mapsto L + 1$. $\square$

The Gordin algebra of this proof is formalized in `DoublingCocycleIdentities`: the halving law $P\psi = \psi/2$ (`log_two_sin_half_add`), the identification $d = 2\psi - \psi \circ T = \log |\tan \pi t|$ (`two_log_two_sin_sub`), and the annihilation $Pd = 0$ (`log_abs_tan_half_add`); with $\int_0^1 d^2 = \pi^2/4$ (`GordinParseval`) this makes every ingredient of the variance formal. [Theorem 9.2](#) itself is not: Mathlib has neither the stationary-ergodic martingale central limit theorem nor its law

of the iterated logarithm, so the step from the exact variance to the constant $\pi/\sqrt{2}$ remains a named import.

Independently of the martingale route, the constant π is excluded by the tail calculus that any LIL proof must supply: with variance $\pi^2/4$, exceedance probabilities at threshold $(\pi - \varepsilon)\sqrt{L \log \log L}$ are $(\log L)^{-(\pi - \varepsilon)^2/(2\sigma^2)} = (\log L)^{-2+o(1)}$ at Gaussian scale, and dyadic block maxima then have summable probabilities, so by Borel–Cantelli the lower bound (1.7) of [1] cannot hold with the constant $\pi - \varepsilon$ infinitely often.

9.3 The source of the error, located in the authors’ own text

Theorem 1.2 of [1] (arXiv numbering: Theorem 2, label `th_theorem2` in the authors’ source file, of which the repository now holds a copy at `docs/papers/arXiv-1502.06738v2/Lacunary_products.tex`) is derived in the paper’s Section 4 from its Lemma 4.1,

$$\limsup_{L \rightarrow \infty} \frac{\sum_{\ell=0}^{L-1} f_1(2^\ell \alpha)}{\sqrt{2L \log \log L}} = \frac{\pi}{\sqrt{2}} \quad \text{a.e.}, \quad f_1(\alpha) = \log |2 \sin \pi \alpha|,$$

which rests on Fortet’s classical sharp LIL (the paper’s Lemma 2.6): $\limsup S_L/\sqrt{2L \log \log L} = \sigma_f$ with σ_f^2 given by the functional (2.9) evaluated in [theorem 9.1](#). The paper evaluates the functional in its equation (4.5):

$$\sigma_{f_1}^2 = \sum_{j=1}^{\infty} \left(\frac{1}{j^2} + 2 \sum_{r=1}^{\infty} \frac{1}{2^r j^2} \right) = \frac{\pi^2}{2}. \quad (4.5)$$

Comparing term by term with the proof of [theorem 9.1](#): the displayed sum is exactly $2c_0 + 2 \sum_r 2c_r$, i.e. each Parseval factor $\int \cos^2 = \frac{1}{2}$ has been replaced by 1. The correct value is $\pi^2/4$; hence Lemma 4.1’s constant is $\pi/2$, and Theorem 1.2’s is $\pi/\sqrt{2}$. Two internal cross-checks against the paper itself: its equation (1.13) states the classical i.i.d. benchmark $\limsup |\sum e^{2\pi i h X_k}|/\sqrt{2N \log \log N} = 1/\sqrt{2}$ *correctly* (variance $\frac{1}{2}$ of a single harmonic retained), and applying the (4.5)-style evaluation to the single harmonic $f = \cos 2\pi \alpha$ would give $\sigma^2 = 1$ and hence a Hadamard-lacunary LIL constant 1 in the $\sqrt{2L \log \log L}$ normalization, contradicting the classical Erdős–Gál value $1/\sqrt{2}$ which Fortet’s formula reproduces. The error is therefore an isolated slip in the evaluation (4.5), identical in the arXiv source and in the published version; it is not an OCR artifact (checked against the publisher’s PDF page images and the authors’ own L^AT_EX source).

9.4 Numerical corroboration

Sampling the doubling map on exact rational orbits $p \mapsto 2p \bmod q$ (odd moduli $q \approx 2^{61}$, uniform residues, so no floating-point orbit corruption at any length) gives

	measured	(8), $\sigma^2 = \pi^2/4$	same with $\sigma^2 = \pi^2/2$
$\text{Var}(S_{20}), 4 \cdot 10^5 \text{ samples}$	45.66	46.06	92.12
$\text{Var}(S_{34}), 4 \cdot 10^5 \text{ samples}$	80.41	80.60	161.20
$\text{Var}(S_{1000}), 4 \cdot 10^6 \text{ samples}$	2459.5	2464.1	4928.2

The covariance $\int_0^1 \psi(x)\psi(2x) dx$ was measured as 0.4112311 against $\pi^2/24 = 0.4112335$. Tail frequencies at $L = 1000$ track the $\sigma = \pi/2$ Gaussian branch from below (the distribution is left-skewed at finite L) and sit far below the $\sigma = \pi/\sqrt{2}$ branch: at threshold $\pi\sqrt{L \log \log L}$ the measured frequency is $5.2 \cdot 10^{-4}$, against $2.7 \cdot 10^{-3}$ for the $\pi/2$ branch and $2.5 \cdot 10^{-2}$ for the branch the printed constant would require. A structural curiosity uncovered along the way: at small L (e.g. $L = 40$) the upper tail is *cut off* by the deterministic Gelfond cap $S_L \leq L \log \sqrt{3} + \mathcal{O}(1)$, which sits below the $c = 2.9$ threshold there; the LIL scale detaches from this cap only for $L \gtrsim 10^2$ – 10^3 , which is why naïve small- L simulations cannot see the asymptotic constant at all, and why the corroboration above was run at $L = 1000$ on exact orbits.

Finding 9.3 (erratum-level correction to [1]). In Aistleitner–Hofer–Larcher, Ann. Inst. Fourier 67 (2017) 637–687: equation (4.5) should read $\sigma_{f_1}^2 = \pi^2/4$; Lemma 4.1 should read $\pi/2$ in place of $\pi/\sqrt{2}$; the constant π in Theorem 1.2 (equations (1.6)–(1.7)) should be $\pi/\sqrt{2} = 2.2214\dots$; and the constant $\pi/\sqrt{\log 2} = 3.7735\dots$ in Theorem 1.1 (equations (1.4)–(1.5)) should be $\pi/\sqrt{2\log 2} = 2.6682\dots$. The paper’s $\sigma_{f_2}^2 = 0$ (equation (4.6)), its Theorem 1.3 (whose exponent involves only λ), and Lemma 5.1’s enclosure for λ are unaffected. The publisher’s PDF (read as page images), the repository’s OCR copy, and the authors’ own arXiv L^AT_EX source (docs/papers/arXiv-1502.06738v2/, where the theorem is numbered Theorem 2, as Document 3 cites it) all show the same formulas, so this is not a transcription artifact of any copy. Consequently Document 3’s Theorem 4.2 constant π must be read as $\pi/\sqrt{2}$, and Document 1’s fluctuation section is correct as written.

Postscript: a corrected edition. Following this audit, the repository owner prepared a corrected critical edition of the paper, stored alongside the OCR transcription as AIF_2017__67_2_637_0-corrected.tex. On the four loci of [finding 9.3](#) it agrees with the audit exactly: (1.4)–(1.5) now read $\pi/\sqrt{2\log 2}$, (1.6)–(1.7) read $\pi/\sqrt{2}$, Lemma 4.1 reads $\pi/2$, and (4.5)–(4.6) carry the Parseval factor $\frac{1}{2}$ (yielding $\pi^2/4$ and 0), with the downstream Section 4 displays updated consistently – checked here by search: no occurrence of the erroneous constants survives anywhere in the edition. The edition goes substantially beyond this audit’s scope: it repairs the near-resonance step (3.8)–(3.10), whose printed form divides by a ceiling that can vanish for admissible indices, via a bipartite maximum-degree-two argument; enlarges Lemma 2.1’s exponential-moment constant from $2\pi^2/3$ to π^2 and propagates the coarse constants $29 \rightarrow 30$, $30 \rightarrow 31$, $85 \rightarrow 88$; removes Theorem 2.5’s inconsistent evenness hypothesis; and corrects a Fourier-tail summation range in Section 4 along with numerous Section 5–6 transcription-level defects. The spot-checks performed for this audit all pass: the internal arithmetic of the new (3.9) ($\|p\|_2^2 \leq \pi^2/12$ on the diagonal, $S_{\ell_1, \ell_2} \leq \pi^2/(2R)$ off it, and $\pi^2/12 + \pi^2/(2(q-1)) \leq \frac{\pi^2}{2} \frac{q}{q-1}$ for $q > 1$), its coherence with the enlarged Lemma 2.1 bound $\exp(\pi^2 \lambda^2 Lq/(q-1))$ (the per-block factor $4\lambda^2 W_i \leq 2\pi^2 \lambda^2 |\Delta_i| \frac{q}{q-1}$ telescopes correctly), and the final Section 4 conversions to $\pi/\sqrt{2\log 2}$. The Section 5–6 repairs were not independently re-derived here and rest on the edition’s own validation.

The meta-lesson for this repository’s methodology deserves a sentence: the error was surfaced not by checking any single document against the literature – Document 3 *matches* the literature – but by insisting that two independently produced documents agree with each other, and then measuring the disputed quantity. Faithful citation is not correctness.

10 Numerical experiments

All experiments evaluate $\log |f|$ by summing $\log |\text{sinc}|$ over levels $h \leq \log_2(\pi x) + 30$ with a quartic Taylor tail (absolute error below 10^{-15}), and integrate over unit lobes by 24-point Gauss–Legendre panels. Scripts: [section B](#).

10.1 Shell means and the constant A_1

$\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |f|/g_1$, by direct quadrature and, independently, as $\varrho_1^{-n} \int_0^1 \mathcal{L}_1^n(W)$ by collocation ($N = 40$) with Clenshaw–Curtis integration:

n	4	6	10	14
direct	0.0912636508	0.0912662060	0.0912661248	0.0912661241
spectral	0.0912636508	0.0912662060	0.0912661248	0.0912661241

The two computations agree to all ten digits at every n (they share no code path beyond the evaluation of H), and the spectral iteration continues to $A_1 = 0.0912661241$ unchanged through $n = 60$. This simultaneously validates the factorization (2), ϱ_1 , κ_1 , and the spectral-gap convergence claimed by Documents 3 and 4.

10.2 Cesàro means: dyadic convergence and the mantissa profile

Running means $\frac{1}{t-t_0} \int_{t_0}^t |f|/(A_1 g_1)$, $t_0 = 8$:

$t = y \cdot 2^n$	$y = 1$	$y = 1.25$	$y = 1.5$	$y = 1.75$
$n = 5$	0.999992	0.964484	1.103575	1.116546
$n = 9$	1.000000	0.971221	1.087225	1.100810
$n = 13$	1.000000	0.971559	1.086373	1.099966
transfer-operator limit	1	0.971581	1.086317	1.099910

The dyadic column locks to 1; the non-dyadic columns converge to the predicted profile values (agreement $2 \cdot 10^{-5}$ to $6 \cdot 10^{-5}$ at $n = 13$). The full profile (figure 1) has range $[0.9249, 1.1133]$ on a step-0.025 grid; exact-point quadrature at $y = 8/7$ and $10/7$ (located by the second wave, confirmed in the second-wave audit) reaches 0.91627 and 1.11814, so the grid under-resolves the singular profile's narrow excursions. This is the numerical content of “(Q3) yes, stabilized (Q2) no”.

10.3 The logarithmic mean

$\frac{1}{\log(t/t_0)} \int_{t_0}^t |f|/g_1 \cdot x^{-1} dx$ at the same mantissas:

	$y = 1$	$y = 1.25$	$y = 1.5$	$y = 1.75$
$n = 7$	0.093861	0.092490	0.095987	0.096493
$n = 13$	0.093861	0.093287	0.094781	0.095032

All columns approach the common constant $A_1^{\log} = 0.09386064$ (the first edition printed the running mean 0.0938610, whose $\mathcal{O}(1/K)$ transient had not died out; corrected — section 13), at the expected $\mathcal{O}(1/\log t)$ rate; the dyadic column is already exact because complete shells contribute equally. This substantiates theorem 5.1(iv).

10.4 Annular maxima and the Lambert formula

Independent maximization (coarse scan over all unit intervals, then bounded refinement), against Document 2's table and its Lambert approximation $\mathcal{A}_k$:

k	$\log M_k$ (this audit)	$x^*/2^k$	Document 2	residual to $\mathcal{A}_k$
3	−11.251778	1.177288	−11.251778	−0.823
6	−25.991031	1.165501	−25.991031	−0.728
9	−46.956338	1.166810	−46.956338	−0.723
12	−74.157742	1.166649	−74.157742	−0.783
13	−84.611550	1.166676	—	−0.814
14	−95.727351	1.083338	—	−0.818
15	−107.493236	1.083331	—	−0.784

Rows 3–12 reproduce Document 2's table to every printed digit. At $k = 14$ the maximizer leaves the mantissa $7/6$ (one doubling to the Gelfond cycle) for $13/12$ (two doublings), exactly the pre-period growth $r = \log_2 k - \log_2 \log k + \mathcal{O}(1)$ predicted by Documents 1 and 2; the Lambert residual stays in a bounded band, consistent with the claimed $\mathcal{O}((\log \log k)^2)$ remainder.

10.5 Two-adic valleys

Peak of $|f|$ on $(2^k, 2^k + 1)$ against $\frac{H(1)^2}{e(k+1)} 2^{-k(k+1)}$ (Documents 2 and 4; $H(1) = 0.5537712759$): log-differences $+0.137$ ($k = 8$), $+0.122$ ($k = 12$), $+0.098$ ($k = 16$), decaying roughly like $1/k$; peak locations 0.8804, 0.9180, 0.9378 against the predicted $1 - 1/(k+1) = 0.8889, 0.9231, 0.9412$. For $m = 3$: differences $+0.156$ ($k = 6$) down to $+0.124$ ($k = 12$), with $|H(3)| = 0.0462022991$. The valley laws are supported as stated (asymptotic, with slow $1/k$ -type approach).

10.6 Variance and tails

See the table in [section 9](#). Method: exact doubling orbits on odd moduli $q \in [2^{61}, 2^{62})$, p uniform on $[1, q)$; each orbit step is one integer shift-and-mod; $\sin$ is evaluated in double precision from the exact rational. This eliminates the two artifacts that invalidate naïve double-precision simulation (orbit collapse after ~ 50 doublings, and spurious exact zeros from short mantissas), both of which were observed and diagnosed before switching to rational orbits.

10.7 The ϱ_1 certificate

Document 4’s Appendix A script was executed verbatim (SymPy 1.14): for both bracketing rational functions the numerator and denominator Sturm root counts on $[0, 1]$ are zero and the sign checks at $t = 0$ pass, certifying $0.66126807 < \varrho_1 < 0.66134891$ in exact arithmetic, hence $2.74801262 < \kappa_1 < 2.74818899$. This is the strongest rigor statement about κ_1 in the corpus, and it is machine-checkable.

The reduction the certificate rests on is formal. CollatzWielandt’s `perron_root_mem_of_t` `wo_sided` proves that any continuous h that is positive on $[0, 1]$ and satisfies $m h \leq \mathcal{L}_1 h \leq M h$ there brackets the growth root, $m \leq \varrho_1 \leq M$ – with no eigenfunction, $\mathcal{L}_1$ being the operator (3) verbatim; and in its abstract form `collatzWielandt_bracket` the argument mentions neither $\mathcal{L}_1$ nor $[0, 1]$ nor sine and cosine, using only monotonicity and positive homogeneity. The numerical half is now formal as well, by a different and Sturm-free route: `PerronRootEnclosure` carries the same cubic test function through the tangent half-angle dictionary $t = \tan(\pi x/4)$ and certifies the two degree-12 integer-polynomial inequalities on $[0, 1]$ with 32-piece subdivided Bernstein certificates (`BernsteinPositivity`), yielding the kernel-checked enclosure $0.66126798 \leq \varrho_1 \leq 0.66134921$. Only the last two digits of the SymPy enclosure above are given up; recovering them needs on the order of two thousand pieces, since the endpoints above approximate the exact range of $\mathcal{L}_1 h/h$ to within 10^{-8} . The κ_1 consequence is now formalized as well, in `KappaOneEnclosure`, and it is precisely the three interval bounds just named that it had to supply. Two of them were new. Mathlib brackets $\log 2$ and $\log 3$ but carries no bound on $\log \pi$; that is proved here to eight decimals,

$$1.1447298855 < \log \pi < 1.1447298860 \quad (\text{Fabius.log_pi_bracket}),$$

by anchoring at the *rational* $c = 3.1415926535$ – so that $3/c = 1 - 283185307/6283185307$ makes the Taylor partial sum of `Real.abs_log_sub_add_sum_range_1e` a numeral – and closing the gap from c to π with `Real.pi_gt_d20` below and `Real.pi_lt_d20` above. Anchoring at $1 - 3/\pi$ directly, the obvious route, is not formalizable at all: that point is irrational, so the partial sum is not a numeral and no `norm_num` can evaluate it. The logarithms at the two rational Perron endpoints follow from $\varrho = \frac{2}{3} \cdot \frac{3}{2} \varrho$, which lets Mathlib’s $\log 2$ carry the scale. Combining them with the dictionary and its antitonicity gives

$$2.7480118 \leq \kappa_1 \leq 2.7481893 \quad (\text{Fabius.kappa_one_enclosure}),$$

so three decimals of κ_1 are kernel-checked. Of the width $1.775 \cdot 10^{-4}$, the ϱ_1 bracket contributes 99.84% and the four logarithm brackets only 0.004%: sharpening κ_1 is entirely a question of sharpening ϱ_1 . The orientation is worth recording, since it is easy to get backwards – the dictionary is antitone, so the *lower* Perron endpoint yields the *upper* bound on κ_1 . The audit’s own $\kappa_1 = 2.74807001487133267 \dots$ lies strictly inside but is not proved.

10.8 Exact spot identities

$I_2(n) = 2^{-n}$ verified to 10^{-12} ($n = 2, 5, 8, 11$); the slow-ray identity $|f(2^k \frac{4}{3})| = |f(\frac{4}{3})|(3\sqrt{3}/(8\pi))^k 2^{-k(k+1)/2}$ verified to 10^{-14} ($k = 6$); Document 4’s peak-ray constant $W_{\kappa_\infty}(2/3) = 0.139129774734829$ reproduced to all digits and its ray identity verified to 10^{-13} ($n = 9$); the collocation eigenvalues $\varrho_2 = \frac{1}{2}$ (exact to 10^{-15}), ϱ_3, ϱ_4 as printed in Document 4.

11 Relation to the repository corpus

11.1 The deleted draft

The *Decay rate conjecture* draft (removed from the working tree; recover with `git show e1cdd6260:".../Decay rate conjecture.claimed`)

$$g(x) \approx x^{\log(2\pi)/\log 2} (\log x)^{-1/2} \exp\left\{\frac{(\log x)^2}{2\log 2}\right\}, \quad g(x)f(x) \sim \text{constant},$$

after an opening that asserts the opposite sign for the logarithmic power. The audit confirms the four documents' unanimous verdict: the pointwise claim is false on three independent grounds (integer zeros; iterated-logarithm fluctuations on typical rays; two-adic valleys), no power of $\log x$ appears in any of the correct mean normalizations, and the draft's power of x is exactly the RMS exponent κ_2 — a meaningful constant attached to the wrong norm. The draft's own derivation sketch (termwise summation of the $\log |\sin|$ Fourier series) is precisely the step that erases the dyadic orbit structure; Documents 3 and 4 show what that structure contributes: the difference between κ_2 and each of κ_1 , κ_0 , κ_∞ .

11.2 Ties to existing material

The Thue–Morse Formula Atlas already contains the sign law $\text{sgn } f|_{(N,N+1)} = \varepsilon_N$, the finite Fourier product, the Gelfond exponent section (the L^∞ row of the present dictionary), and—as the structural twin of the universal factor—the boundary flatness theorem $\log E(t) = -\log^2(1/t)/(2\log 2) + \mathcal{O}(\log(1/t))$ for $E(t) = \prod_j (1 - e^{-2^j t})$, including the remark that a bounded log-periodic fluctuation persists at the next order. The mantissa profile of [figure 1](#) is the exact analogue of that fluctuation for the decay problem, with the new feature that here the fluctuation is driven by a *singular* measure. On the Lean side, `FourierProduct/PoissonSummation` already carry the product, the scaling law, and rapid decay; `FabiusLogSquaredAsymptotic` and its neighbors carry the small-argument $-\log^2$ layer. The decay layer proper is by now substantially covered as well: the shell factorization and the κ -algebra (`SincProductShells`); the sharp Gelfond bound and both mean brackets (`SharpGelfondBound`, `SincMeanBracket`, `LacunaryMeanSharp`); the exact variance package (`BirkhoffVariance`, `BirkhoffProductBridge`, `CovarianceLadder`, `CocycleParseval`, `GordinParseval`); the lobe geometry — zero orders, one peak, Thue–Morse sign (`PhiZeroOrder`, `OnePeakPerLobe`, `RvachevProductContinuity`, `ThueMorseLobeSign`); the exponent dictionary (`KappaDictionary`) and the certified enclosures of all four exponents (`KappaOneEnclosure`, `KappaConstantEnclosures`); the two-sided κ_∞ envelope (`GlobalDecayEnvelope`, `PeakRayEnvelope`); and the hypothesis-carrying skeletons of the mean theorems (`ShellLinearity`, `BoundednessSkeleton`, `CesaroProfileSkeleton`, `LogMeanSkeleton`). The gap between that list and this document is the RPF analytic block and the martingale limit theorems, exactly as [section 5.2](#) predicts.

11.3 Formalization targets, ordered by feasibility

1. The exact factorization (2) and the κ -algebra: finite manipulation of already-formalized objects. **Done**, in more generality: `GeometricScaleProducts` proves the renormalization law for geometric-scale infinite products $\prod_n g(z/c^n)$ over any field into any commutative topological monoid (`tprod_geom_scale`, `tprod_geom_scale_pow`, induction-free), together with the monoid-valued tail-multipliability direction missing from `Mathlib` (`Multipliable.of_nat_add`); `SincProductShells` specializes to the standalone sinc product (`rvachevFourierProduct_two_pow_mul`), proves the modulus shell factorization with the triangular power (`norm_rvachevFourierProduct_two_pow_mul`, `prod_range_two_pow_succ`), and formalizes the κ -algebra as `shell_exponent_identity`.

2. $I_2(n) = 2^{-n}$: finite Parseval over the Thue–Morse block, adjacent to ThueMorseBlockAlgebra. **Done**, in more generality than stated here: the module LacunaryRieszIntegral proves the master orthogonality theorem `integral_cos_mul_rieszProduct` for arbitrary gap-2 integer frequencies, amplitudes, and phases (every Fourier coefficient of a lacunary Riesz product below the spectral floor), with the identity of [proposition 5.4](#) as the corollary `integral_prod_sin_sq_two_pow` via arbitrary bases (`integral_prod_sin_sq_pow`).
3. The Gelfond bound via Document 4’s two-step inequality (a single polynomial inequality, `nlinarith` territory) or Document 2’s subaction identity (pure algebra). **Done**, in more generality: `GelfondLogisticBound` proves the product bound along *every* orbit of the full logistic map (`logistic_prod_le`), specializes to the dyadic sine products (`prod_sin_sq_two_pow_le`, `abs_prod_sin_two_pow_le`) with sharpness at $t = 1/3$ (`prod_sin_sq_two_pow_third`), and combines with the L^2 identity to formalize the lower bracket of [proposition 5.3](#) ($I_1(n) \geq \sqrt{3/5} 3^{-n/2}$, hence $\varrho_1 \geq 3^{-1/2} > \frac{1}{2}$) as `le_mul_integral_prod_abs_sin_two_pow` – the elementary input to the singularity argument of [theorem 5.5](#).
4. The ϱ_1 enclosure via the Sturm certificate: rational positivity on an interval; decidable, and the certificate is already machine-readable. **Done**, at marginally wider constants and with no Sturm module at all: `PerronRootEnclosure` exhibits Document 4’s optimized cubic test function and certifies

$$0.66126798 \leq \varrho_1 \leq 0.66134921$$

(`perron_root_enclosure`, `exists_perron_root_enclosure`) by discharging the two positivity checks with 32-piece subdivided Bernstein certificates – exact integer coefficients, verified by ring normalization and literal sign checks alone (`BernsteinPositivity`, `nonneg_on_Icc_of_pieces`). The reduction underneath is the previously formal `perron_root_mem_of_two_sided`. The SymPy enclosure of [section 10.7](#) is $\approx 10^{-7}$ sharper; its endpoints sit within 10^{-8} of the exact range of $\mathcal{L}_1 h/h$, so reaching them by subdivision needs roughly two thousand pieces (the method converges quadratically in the piece width) – a mechanical, if bulky, sharpening.

5. The valley asymptotics (elementary boundary-maximum analysis). **The local half is done**: the zero order at m is exactly $1 + v_2(m)$ with an explicit positive constant (`LatticeFiber`, `ZeroOrderTail`, `PhiZeroOrder`), and the two extremes – simple zeros at odd integers, order $j + 1$ at 2^j , hence unbounded order along the dyadic integers – are `ValleyZeroOrders`. The asymptotic constant of the valley law $E_{m2^k} \sim \frac{H(1)|H(m)|}{e^{(k+1)}} (m2^k)^{-(k+1)}$ is not formalized.
6. The variance $\sigma^2 = \pi^2/4$: Parseval for the Clausen series of $\log |2 \sin \pi x|$ plus the geometric covariance sum; would also formally certify [finding 9.3](#). **Done**, in the exact finite- m form of [theorem 9.1](#) rather than only in the limit: `BirkhoffVariance` (`integral_sq_birkhoff_sum_all`) and `BirkhoffProductBridge` (`integral_sq_log_prod_two_sin_all`) prove $\int_0^1 S_m^2 = \frac{\pi^2}{4} m - \frac{\pi^2}{3} (1 - 2^{-m})$ for every m , including $m = 0$, on the covariance ladder `CovarianceLadder` and the two Clausen values `CocycleParseval`, `GordinParseval`, with the Gordin algebra in `DoublingCocycleIdentities`. This item was still listed as an open target in earlier editions of this audit.
7. Hardest: Birkhoff/ergodic layers and the Perron–Frobenius gap; import as named hypotheses first, as Documents 3 and 4 recommend. That recommendation has been carried out for the mean theorems: the limit-theoretic cores of [theorem 5.1\(i\),\(iii\)](#) and [theorem 5.7](#) are formal for an arbitrary cumulative mass with the RPF layer left as a hypothesis (`BoundednessSkeleton`, `CesaroProfileSkeleton`, `LogMeanSkeleton`, `GeometricCesaro`), as is [lemma 5.2](#) (`ShellLinearity`). On the operator side, existence of the growth root $\varrho_1 \in [\frac{1}{2}, \sqrt{2}/2]$ by Fekete (`PerronRootExistence`) and the Lasota–Yorke pair (`TransferOperatorStep`,

LasotaYorkeIteration) are formal; the spectral gap, the eigenfunction, the eigenmeasure and its singularity, and the martingale limit theorems are not, and remain the frontier.

12 Verdict

1. **On the question asked.** The bounded form (Q3) is solved, constructively: $g = A_1 g_1$ of (4). The exact-limit form (Q2) is unattainable for any gauge in the natural (shell-shape-stabilizing, in particular Hardy-field) class; it becomes attainable along dyadic times, or exactly and for all times under the logarithmic mean — and, as the second wave later proved, exactly and for all times under the ordinary mean too, by a necessarily shell-adaptive analytic gauge outside the natural class (section 13). This composite answer — not any single formula — is the correct resolution of [6]; within it, the 2018 answer of Negometyanov (a truncated q -Pochhammer representation, useful for bounded-range evaluation) contains no asymptotic law, as the question’s author already noted at the time. The two readings the four documents left underdeveloped are now treated in full: the lobe-maxima envelope decomposes into the same four regimes and rejoins the κ_1 gauge through the peak–integral comparison, with $\Theta_{\text{peak}} = 1.9237$ tying the peak and integral readings together (section 6); and the root-mean-square reading is exactly solvable — closed subleading eigenvalue $-\frac{1}{4}$, alternating $\mathcal{O}(1/x)$ convergence, total-energy identity — yet still carries the dyadic phase obstruction (section 7).
2. **On the four documents.** All are technically serious, and their factual overlap survived every numerical test applied here, in several cases to all printed digits. Documents 3 and 4 answer the question; Document 4 is the most rigorous where they overlap (its ϱ_1 certificate passes exact-arithmetic execution); Document 2 is the sharpest on the extremal geometry (its numerics reproduce exactly) and the only one to scope the problem honestly; Document 1 is subsumed by Document 2 except for its fluctuation section — which turned out to be the corpus’s single most valuable page.
3. **Errors found.** One substantive: the constant π in Document 3’s Theorem 4.2, inherited verbatim from Theorem 1.2 of the published [1], whose own equation (4.5) drops a Parseval factor $\frac{1}{2}$; the correct constant is $\pi/\sqrt{2}$ (finding 9.3), as Document 1 states. All other deviations found were of scope, convergence-speed candor, or single-ulp arithmetic.
4. **Recommended dispositions.** Absorb into the frontier volume: the κ_q dictionary with the certificate-backed $q = 1$ row; the two-part answer to (Q2)/(Q3) with the mantissa profile; the corrected fluctuation constants; the Lambert-form annular envelope and valley laws with explicit convergence-speed caveats. The OCR copy of [1] is annotated at (1.4)–(1.7), Lemma 4.1 and (4.5) per the repository’s policy on evident errors in stored papers, and the same directory now carries a corrected critical edition of the paper whose constants agree with finding 9.3 and whose further repairs are catalogued in the postscript of section 9; communicating the correction to the authors remains recommended. Retire the four drafts after absorption, per the inbox policy; the accessible companion (Rvachev_Up_Fourier_Decay_Gentle_Guide/) should survive absorption as the entry point for readers.

13 Postscript: the second wave (27 August 2026)

After this audit was written, four further documents (rvachev_up_fourier_decay-5 through -8), each composed with access to it, arrived and were audited separately in `./Rvachev_Up_Fourier_Decay_Second_Wave_Audit/`. Their net effect on this document is recorded here; the in-text corrections above are marked with references to this section.

1. **The unrestricted (Q2) is attainable.** Documents 7 and 8 prove, by two independent constructions (dyadic mass equalization on slowly refining partitions, riding the within-shell baseline slope $-(k + \kappa_1 + 1)/2$; then Hoischen–Carleman entire approximation of the gauge, or of its reciprocal, together with its derivative), that a positive, real-analytic, strictly decreasing gauge attains (Q2) for all t , with $\log g = -\log^2 x / (2 \log 2) - \kappa_1 \log x + o(\log x)$. Document 7 also strengthens [theorem 5.5](#) from stabilizing shapes to all bounded-distortion gauges. [Theorem 5.5](#) itself stands; this audit’s summary slogan “unattainable in principle” did not, and has been corrected in place. Any exact gauge necessarily develops unbounded, ever finer within-shell structure; no stabilized formula works.
2. **Profile range.** The step-0.025 grid underlying [figure 1](#) misses narrow excursions of the singular-continuous profile near $y = 8/7$ and $y = 10/7$; exact-point quadrature reaches $\mathcal{P} \approx 0.91627$ and 1.11814 (values confirmed independently at levels 17–19 in the second-wave audit, which also reproduces Document 6’s level-18 table to $4 \cdot 10^{-10}$ and Document 7’s level-19 extremes to all printed digits). The figure’s curve is unchanged – it is correct for its grid – but its range is a lower bound, not the span.
3. **Constants.** Forty-digit collocation gives $\varrho_1 = 0.66132260206056561507\dots$, $A_1 = 0.09126612413158821229\dots$, $A_1^{\log} = 0.09386063575678126354\dots$ (this audit’s first edition printed $A_1^{\log} = 0.0938610$, a finite-shell running mean wrong in the seventh figure – corrected above), and the resolved $\kappa_1 = 2.74807001487133267\dots$ (the widely copied sixteen-digit print 2.748070014871334 inherits the fifteen-digit truncation of ϱ_1 ; this audit’s constants table had bracketed the uncertainty correctly, its abstract had not). New exact-ray constant: $|f(2^n \cdot \frac{2}{3})| = C_* E_{\kappa_\infty}(2^n \cdot \frac{2}{3})$ for all $n \geq 0$ with $C_* = 0.13912977473482934529\dots$ (Document 6; verified to 38 digits).
4. **Fluctuation theory.** Document 6’s one-line Gordin decomposition ($\mathcal{P}\psi = \psi/2$, $d = 2\psi - \psi \circ T = \log |\tan \pi t|$, $\int d^2 = \pi^2/4$ exactly) gives the shortest self-contained proof of [theorems 9.1](#) and [9.2](#); the second wave unanimously confirms [finding 9.3](#).

A Constants

symbol	value	provenance
$1/(2 \log 2)$	0.72134752...	universal quadratic coefficient
κ_0	3.151496129472319	$\frac{3}{2} + \log_2 \pi$ (exact form)
κ_1	2.74807001487133267	$\frac{1}{2} + \log_2(\pi/\varrho_1)$; resolved in the second wave
κ_2	2.651496129472318	$\log_2(2\pi)$ (exact form)
κ_∞	2.359014879111741	$\frac{3}{2} + \log_2(\pi/\sqrt{3})$ (exact form)
ϱ_1	0.66132260206056561507	collocation (second-wave precision); certified in (0.66126807, 0.66132260206056561507)
ϱ_2	1/2	exact
ϱ_3	0.394896632370234	collocation
ϱ_4	0.320194101601104	collocation
A_1	0.091266124131588212	shell-mean limit (direct = spectral)
$A_1^{\log}$	0.093860635756781264	logarithmic-shell limit (corrects the first edition's 0.0938610)
C_*	0.139129774734829345	exact peak-ray constant $ f(2^n \cdot \frac{2}{3}) /E_{\kappa_\infty}$, second wave
A_2	0.103008918	RMS shell constant, $A_2^2 = 0.0106108372$
$\lambda_2(\mathcal{L}_2)$	-1/4 (exact, double)	proposition 7.1 ; rest of spectrum $< 3 \cdot 10^{-4}$
$\lambda_2(\mathcal{L}_1)$	$-0.13435 \pm 0.0500 i$	collocation; $ \lambda_2 /\varrho_1 = 0.217$
Θ_{peak}	1.92372(3)	$\sum_{\text{shell}} E_n / \int_{\text{shell}} f $, $k = 9-14$
A_1^{peak}	0.175569	$\Theta_{\text{peak}} A_1$ (conjectural limit)
$\int_{\mathbb{R}} f^2$	0.808873535967972	two independent computations, proposition 7.3
L^2 profile range	[0.894, 1.192]	step-0.025 grid, $n = 13$
profile range	$\supseteq [0.91627, 1.11814]$	narrow features at $y = 8/7, 10/7$ (second wave); 0.025 grid sees [0.91627, 1.11814]
$H(1)$	0.5537712758881009534	30-digit product
$ H(3) $	0.0462022991	direct product
σ^2 (per step)	$\pi^2/4 = 2.4674011$	proved (theorem 9.1); measured 2.4595 at $L = 10^3$
LIL constant	$\pi/\sqrt{2} = 2.2214415$	corrected; printed value π

B Verification scripts

The complete scripts (stage1.py-stage6b.py, in `verification_scripts/`) are short, self-contained NumPy/SymPy programs; stage6.py/stage6b.py produce the lobe-maxima statistics, the $\mathcal{L}_2$ spectrum, the RMS constants and profile, the Plancherel check, and the κ_q table of [sections 6](#) and [7](#). The essential kernels follow.

Listing 1: Evaluation of $\log |f|$ and the exact-orbit variance measurement

```
def log_abs_f(x):
    # sum over levels h with quartic tail
    H = ceil(log2(pi*max(x))) + 30
    return sum(log|sinc(pi*x/2**h)| for h in 0..H) # |u|<1e-4: -u^2/6-u^4/180

# variance of S_L on exact rational orbits (no float orbit error):
q = 2*randint(2**60, 2**61) + 1 # odd modulus ~2^61, per sample
p = randint(1, q) # uniform residue
for j in range(L):
    S += log|2 sin(pi*p/q)|
    p = (2*p) % q # exact doubling
# Var(S_1000) = 2459.5 vs 1000*pi^2/4 - pi^2/3 = 2464.1
```

Listing 2: Transfer operator: Perron root, shell constant, and mantissa profile

```
x[j] = (1 - cos(j*pi/N))/2 # Chebyshev-Lobatto on [0,1]
L1 = 0.5*( diag(sin(pi*x/2)) @ Bary(x -> x/2)
          + diag(cos(pi*x/2)) @ Bary(x -> (x+1)/2) )
```

```
rho1 = leading_eigenvalue(L1)          # 0.661322602060565
A1   = lim_n rho1**(-n) * clenshaw_curtis( L1**n @ W )   # 0.0912661241
F(z) = lim_n rho1**(-n) * int_0^z W*P_n / (same at z=1) # profile mass
P(y) = (1 + F(y-1)) / y                # Cesaro limit at t = 2^k * y
```

Listing 3: Document 4’s certificate, as executed (passes)

```
F(z) = 1 + 376189/10^6 z - 69093/10^6 z^2 + 15483/10^6 z^3
R(t) = (u F(u) + v F(v)) / (2 F(2uv)), u = 2t/(1+t^2), v = (1-t^2)/(1+t^2)
for expr in (R - 0.66126807, 0.66134891 - R):
    assert numerator, denominator have no roots on [0,1] # Sturm
    assert positive at t = 0
# certified: 0.66126807 < rho_1 < 0.66134891
```

References

- [1] C. Aistleitner, R. Hofer, and G. Larcher, *On evil Kronecker sequences and lacunary trigonometric products*, Ann. Inst. Fourier (Grenoble) **67** (2017), no. 2, 637–687; arXiv:1502.06738. Local copies: Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/ (published version, OCR, with editorial notes; the same directory holds the corrected critical edition AIF_2017__67_2_637_0-corrected.tex) and Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/ (authors’ $\text{\LaTeX}$ source).
- [2] J. Arias de Reyna, *An infinitely differentiable function with compact support: definition and properties*, arXiv:1702.05442; original 1982.
- [3] É. Fouvry and C. Mauduit, *Sommes des chiffres et nombres presque premiers*, Math. Ann. **305** (1996), 571–599.
- [4] R. Fortet, as cited in [1], Lemma 2.6 (sharp LIL for $\sum f(2^\ell \alpha)$ with Hölder f).
- [5] A. O. Gel’fond, *Sur les nombres qui ont des propriétés additives et multiplicatives données*, Acta Arith. **13** (1967/68), 259–265.
- [6] V. Reshetnikov, *Asymptotic decay rate of an infinite product of sinc functions*, Mathematics Stack Exchange question 2745497 (April 2018); OCR copy in drafts/rvachev_up_fourier_decay/.
- [7] V. Reshetnikov, *Extrema of an infinite product of sinc functions*, Mathematics Stack Exchange question 2742261 (2018).